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# Variational Bayesian Message Passing Receiver for Uplink ISAC Systems

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***Abstract*—Uplink integrated sensing and communication (ISAC) systems enable base stations to jointly estimate radar target parameters and detect communication signals from multiple users using shared time-frequency resources. However, the mutual interference between radar echoes and communication signals poses significant challenges for signal processing, particularly under practical conditions such as comparable power levels and uncertain noise variance. To address these challenges, we propose two variational Bayesian message passing receivers based on mean-field and Bethe approximations. The mean-field method derives a closed-form expression for the variational free energy and incorporates a noise variance learning mechanism to improve convergence. The Bethe-based method further refines the posterior approximation by analyzing the stationary points of the Bethe free energy and includes a fixed-point update for noise variance. We also develop a state evolution (SE) analysis to characterize the convergence behavior and estimation accuracy of the Bethe method. Simulation results demonstrate that both methods outperform baseline schemes in terms of mean squared error (MSE) and bit error rate (BER), while maintaining similar computational complexity.**

***Index Terms*—Integrated sensing and communication, variational Bayesian, free energy, state evolution.**

## I. Introduction

In next-generation wireless networks, pivotal applications such as the Internet of Things (IoT), smart cities, intelligent factories, and unmanned aerial vehicles (UAVs) are increasingly integrating sensing functionalities into traditional communication frameworks. To meet the dual demands of high sensing accuracy and reliable communication while efficiently utilizing spectral and hardware resources, integrated sensing and communication (ISAC) has become a key enabling technology for 6G. ISAC leverages shared radio-frequency spectrum and hardware to perform both tasks simultaneously, offering significant gains in resource efficiency for both hardware and spectrum [1], [2].

Theoretical studies have explored the fundamental trade-offs between sensing and communication (S&C), laying a strong foundation for ISAC system design. For instance, [3] characterizes the S&C performance region via the Cramér-Rao Bound (CRB) and communication capacity, revealing the trade-off in degrees of freedom (DoFs) between S&C. Other works, such as [4], [5], adopt mutual information (MI) as a unified metric to analyze this trade-off. These studies collectively demonstrate that co-designed ISAC systems can significantly outperform systems with separate sensing and communication.

While theoretical analyses provide critical performance bounds and trade-offs, the practical realization of ISAC systems relies on effective transceiver design. Existing works have primarily concentrated on transmitter-side optimization, such as beamforming [6], [7] and waveform design [8], [9] for the downlink phase. In contrast, the receiver-side signal processing in the uplink phase has received comparatively less attention. Yet, uplink processing is equally vital. By leveraging the spatial resources provided by massive multiple-input multiple-output (MIMO), base stations (BSs) must simultaneously detect communication signals transmitted by multiple users and estimate sensing target parameters from radar echo signals [10]. This task is complicated due to mutual interference between the two types of signals, both occupying the same time-frequency resources, and requires algorithms that jointly optimize communication accuracy (e.g., bit error rate, BER) and sensing performance (e.g., mean squared error, MSE).

Earlier research on uplink ISAC receivers addresses this challenge using interference cancellation (IC) strategies. In the scenario considered by [10], the radar echo signals and communication signals overlap only during certain time snapshots. Their method first estimates the target angle parameters using the full coherence block, then uses non-overlapping radar echoes to estimate the reflection coefficients. The radar echo component is subtracted from the received signal before applying a zero-forcing (ZF) detector for communication signal recovery. A similar sequential IC strategy is proposed in [11], which first detects communication signals using ZF, subtracts the estimated communication component, and then estimates the sensing target parameters. Alternatively, [12] employs a sub-optimal IC-based architecture where a maximum likelihood (ML) detector is used for communication signal detection, followed by minimum mean squared error (MMSE) estimation of radar parameters.

As noted in [12], [13], IC-based receivers suffer substantial performance degradation when radar and communication signals have comparable power levels. Additionally, their sequential processing mechanism can lead to error propagation. To address these limitations, a non-IC method is developed from an optimal Bayesian perspective for a simplified scenario [12]. In addition, [13] introduces a projection-based scheme

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that mitigates sensing interference during communication signal detection. Compared with IC-based methods, projection-based receivers do not rely on power asymmetry and offer improved S&C performance. However, such schemes are often computationally prohibitive and fail to exploit the statistical information of the radar parameters.

To mitigate these drawbacks, several alternating optimization methods have been investigated in ISAC systems [14], [15]. These methods iteratively perform radar parameter estimation and communication signal detection to improve joint performance. In this context, several variational inference (VI)-based frameworks have been proposed to leverage the statistical correlations among various types of parameters and signals, thereby enabling joint optimization of the ISAC systems [16]–[18]. For instance, [16] introduces an iterative Bayesian receiver for joint localization and sensing, while [17] extends VI to a multi-BS orthogonal time-frequency space (OTFS)-based ISAC systems to perform joint estimation of communication data and channel parameters. Additionally, [18] develops an expectation-maximization (EM) turbo bilinear subspace VI algorithm for joint environment sensing and data detection. These works demonstrate VI's potential in handling statistical inference involving several classes of variables.

However, existing VI-based methods for ISAC systems often rely on intuitive factor graphs based on probabilistic relationships among various variables, which may limit generality and scalability. In this work, we develop message passing algorithms through optimizing the variational free energy, which offers a principled route for Kullback-Leibler (KL) divergence minimization [19]. Based on this framework, we derive closed-form expressions for different variational free energies to develop corresponding message passing rules and noise variance learning methods. Unlike prior works such as [17], [20], which assume predefined noise distributions, we treat the noise variance as a hyperparameter and analytically derive update rules by locating stationary points of the free energy. Moreover, current VI-based approaches for ISAC systems generally lack theoretical convergence analysis. To fill this gap, we establish a state evolution (SE)-based framework that characterizes the behavior of the proposed message passing algorithm in the large-system limit. In terms of system modeling, our work extends the setup in [12] by generalizing from a single-user, single-target scenario to a more practical multi-user, multi-target uplink ISAC configuration.

The specific contributions of this paper are listed as follows:

- **Mean-field method with noise variance learning:** We develop a variational message passing algorithm based on the mean-field approximation for joint estimation of sensing parameters and detection of communication signals in an uplink ISAC system. The closed-form expression of the variational free energy is derived to enable efficient noise variance learning, further accelerating the algorithmic convergence.
- **Bethe method for refined posterior approximation:** We extend the framework to Bethe variational free energy, providing a more accurate approximation. Based on this, we formulate approximate message passing equations and derive a fixed-point formula for noise variance update by investigating the stationary points of Bethe free energy.
- **SE-based performance analysis:** The convergence properties of the proposed Bethe method are analyzed through SE equations, which characterize the convergence behavior and verify the performance gains achieved through noise variance learning.
- **Effectiveness validation:** Simulation results and complexity analysis show that the proposed mean-field and Bethe methods outperform existing linear, IC-based, and iterative alternating optimization baselines, with comparable computational complexity.

The remainder of the paper is organized as follows. Section II introduces the uplink ISAC system model, with the problem formulation in Section III-A. Section III-B reviews relevant variational Bayesian methods, and Section IV presents the proposed mean-field and Bethe-based algorithms. Section V provides the state evolution analysis, and Section VI reports numerical results and complexity comparisons. Finally, Section VII concludes the paper.

*Notation*: Bold uppercase, bold lowercase, and lowercase letters denote matrices, vectors, and scalars, respectively. The notations $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^{-1}$ represent the transpose, the conjugate transpose, and the inversion of a matrix, respectively. $\mathbf{I}$ denotes the identity matrix, and $\mathbb{I}$ is the indicator function. We use $\|\cdot\|$, $\mathrm{vec}(\cdot)$, and $\mathrm{Tr}(\cdot)$ to denote the $l_2$ norm, vectorization and trace of a matrix. $\otimes$ is the Kronecker product operation. $\Re\{\cdot\}$ denotes the real part operator. $\langle\cdot\rangle$ represents the arithmetic mean of all elements of a matrix. $\mathcal{N}_{\mathbb{C}}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a multi-variate complex Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. $\mathcal{U}(\mathcal{X})$ denotes a uniform distribution defined on a discrete set $\mathcal{X}$. The statistical expectation operation with respect to distribution $p$ is represented by $\mathbb{E}_p[\cdot]$. Table I summarizes the key notations used in this paper. Furthermore, we define a normalization function $Z(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with respect to distribution $p(\mathbf{x})$ as

$$Z(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \int \mathrm{d}\boldsymbol{x} p(\boldsymbol{x}) \exp\{-(\boldsymbol{x}^H - \boldsymbol{\mu}^H)\boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\}. \quad (1)$$

Then two mean and variance functions with scalar parameters $\mu$ and $\sigma^2$ with respect to variable $x$ and distribution $p(x)$ are defined as

$$F_a(\mu, \sigma^2) = \int \mathrm{d}x \frac{x}{Z(\mu, \sigma^2)} p(x) \exp\{-\frac{|x-\mu|^2}{\sigma^2}\}, \quad (2)$$

$$F_c(\mu, \sigma^2) = \int \mathrm{d}x \frac{|x - F_a(\mu, \sigma^2)|^2}{Z(\mu, \sigma^2)} p(x) \exp\{-\frac{|x-\mu|^2}{\sigma^2}\}. \quad (3)$$

## II. System Model

As is shown in Fig. 1, we consider a MIMO ISAC system sensing $L$ targets while serving $K$ single-antenna user equipment (UE) devices. The ISAC BS is equipped with uniform linear arrays (ULA), comprising $N_t$ transmit antennas and $N_r$ receive antennas. The ISAC system operates in a synchronized time-division duplex (TDD) mode, encompassing three stages: target searching, target tracking/downlink communication, and target tracking/uplink communication. During the target searching stage, the BS transmits downlink pilots to

TABLE I
SUMMARY OF KEY NOTATIONS

| Notations | Meaning |
|---|---|
| $N_t, N_r$ | transmit/receive antenna number of the ISAC BS |
| $L$ | number of sensing targets |
| $K$ | number of users |
| $S$ | number of snapshots |
| $M$ | modulation order |
| $T$ | iteration number |
| $\mathcal{X}$ | modulation constellation set |
| $N_0$ | noise variance |
| $\delta$ | learnable noise variance |
| $\zeta_R, \zeta_T$ | correlation coefficients of communication channel |

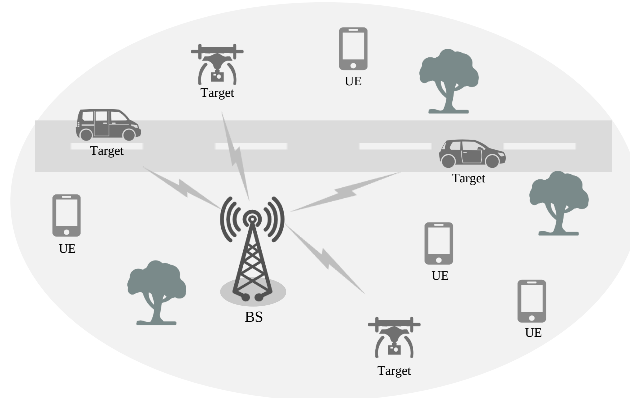



Fig. 1. The uplink ISAC scenario.

identify and locate the targets of interest and performs an initial estimation of radar parameters. In the second stage, the BS emits dual-functional waveform for concurrently executing target tracking and downlink communication tasks. The third stage commences with the BS conducting channel estimation (CE) to acquire the communication channel state information (CSI) by utilizing pilot signals from users. No sensing tasks are performed during the channel estimation step. Subsequently, the BS sends sensing signals to the targets, while the UEs transmit uplink signals to communicate with the BS simultaneously. Consequently, the received signal at the BS comprises a combination of target echoes, uplink signals, and noise. This paper primarily focuses on the joint target tracking and uplink communication stage. The corresponding radar model, uplink communication model, and self-interference model are elaborated in the following subsections.

### A. Radar Model

Let $d$ and $\lambda$ denote the antenna spacing and the wavelength, and the steering vector with $N$ antennas at angle $\theta$ can be expressed as

$$\mathbf{a}\left(N,\theta\right)=\left[1,e^{j\frac{2\pi}{\lambda}d\sin(\theta)},\ldots,e^{j\frac{2\pi}{\lambda}d(N-1)\sin(\theta)}\right]^{T}\in\mathbb{C}^{N}. \tag{4}$$

Denote $\mathbf{r}[s]\in\mathbb{C}^{N_t}$ as the probing signal vector sent from the BS during the $s$-th snapshot, which satisfies $\mathbb{E}[\mathbf{r}[s]]=\mathbf{0}$ and the power constraint $\mathbb{E}[\mathrm{Tr}(\mathbf{r}[s]\mathbf{r}[s]^H)]\le N_tP_r$. Hence, the radar echo received by the BS at the $s$-th snapshot $\mathbf{y}_{\mathrm{r}}[s]\in\mathbb{C}^{N_r}$ is given by

$$\mathbf{y}_{\mathrm{r}}[s]=\sum_{l=1}^{L}\alpha_l\mathbf{a}\left(N_r,\theta_l\right)\mathbf{a}^H\left(N_t,\theta_l\right)\mathbf{r}[s]+\mathbf{n}_{\mathrm{r}}[s], \tag{5}$$

where $\mathbf{n}_{\mathrm{r}}[s]\sim\mathcal{N}_{\mathbb{C}}(\mathbf{0},\frac{N_0}{2}\mathbf{I}_{N_r})$ is the additive white Gaussian noise (AWGN), $\alpha_l\in\mathbb{C}$ and $\theta_l$ represent the reflection coefficient and the azimuth angle of the $l$-th target, respectively. According to [21], we assume the normalized reflection coefficient $\alpha_l$ is modeled as a complex Gaussian distribution $\mathcal{N}_{\mathbb{C}}(\mu_l,1)$, where the non-zero mean $\mu_l$ is determined by the distance between the $l$-th target and the BS. With a more compact form, the received signal matrix within a coherence block ($S$ snapshots) can be rewritten as

$$\mathbf{Y}_{\mathrm{r}}=\mathbf{A}\left(N_r,\Theta\right)\mathrm{diag}\left(\boldsymbol{\alpha}\right)\mathbf{A}^H\left(N_t,\Theta\right)\mathbf{R}+\mathbf{N}_{\mathrm{r}}, \tag{6}$$

where $\mathbf{Y}_{\mathrm{r}}=[\mathbf{y}_{\mathrm{r}}[1],...,\mathbf{y}_{\mathrm{r}}[S]]\in\mathbb{C}^{N_r\times S}$, $\mathbf{R}=[\mathbf{r}[1],...,\mathbf{r}[S]]\in\mathbb{C}^{N_t\times S}$, $\mathbf{N}_{\mathrm{r}}=[\mathbf{n}_{\mathrm{r}}[1],...,\mathbf{n}_{\mathrm{r}}[S]]\in\mathbb{C}^{N_r\times S}$, $\mathbf{A}\left(N,\Theta\right)=[\mathbf{a}\left(N,\theta_1\right),...,\mathbf{a}\left(N,\theta_L\right)]\in\mathbb{C}^{N\times L}$ with $\Theta=[\theta_1,...,\theta_L]$, and $\boldsymbol{\alpha}=[\alpha_1,...,\alpha_L]^T\in\mathbb{C}^L$.

### B. Uplink Communication Model

During the uplink communication stage, the information bits are mapped to transmitted symbol matrix $\mathbf{X}=[\mathbf{x}[1],...,\mathbf{x}[S]]\in\mathcal{X}^{K\times S}$ using $M$-QAM modulation, where $\mathcal{X}$ is the modulation constellation set. The signal power of each symbol is denoted by $E_s$. The received signal $\mathbf{Y}_{\mathrm{c}}=[\mathbf{y}_{\mathrm{c}}[1],...,\mathbf{y}_{\mathrm{c}}[S]]\in\mathbb{C}^{N_r\times S}$ at the BS can be formulated as

$$\mathbf{Y}_{\mathrm{c}}=\mathbf{H}_{\mathrm{c}}\mathbf{X}+\mathbf{N}_{\mathrm{c}}, \tag{7}$$

where the AWGN matrix $\mathbf{N}_{\mathrm{c}}\in\mathbb{C}^{N_r\times S}$ follows the same distribution as $\mathbf{N}_{\mathrm{r}}$. Here, we consider a correlated MIMO channel between the UEs and the BS by the Kronecker model [22]. The channel matrix is given by $\mathbf{H}_{\mathrm{c}}=\mathbf{R}_r^{1/2}\mathbf{T}\mathbf{R}_t^{1/2}$, where $\mathbf{T}$ represents the Rayleigh matrix with i.i.d. entries which obey a complex Gaussian distribution $\mathcal{N}_{\mathbb{C}}(0,1)$. The correlation matrices of the receiver and transmitter sides are defined as

$$\mathbf{R}_{r_{N_r,K}}=\begin{cases}\left(\zeta_R e^{j\phi}\right)^{N_r-K}, & N_r\le K,\\ \mathbf{R}^*_{r_{N_r,K}}, & N_r>K;\end{cases} \tag{8a}$$

$$\mathbf{R}_{t_{N_r,K}}=\begin{cases}\left(\zeta_T e^{j\phi}\right)^{N_r-K}, & N_r\le K,\\ \mathbf{R}^*_{t_{N_r,K}}, & N_r>K,\end{cases} \tag{8b}$$

where $\zeta_R$ and $\zeta_T$ denotes the correlation coefficients, and $\phi$ is the given phase.

### C. Self-Interference Model

In the third stage, the BS transmits radar probing signals through its transmit antennas while simultaneously receiving radar echo and communication signals using its receive antenna array. This configuration inherently leads to self-interference caused by the transmitted sensing signals. The self-interference channel between the BS's transmit and receive antennas is represented as $\mathbf{H}_{\mathrm{SI}}\in\mathbb{C}^{N_r\times N_t}$. Therefore, the received self-interference signal can be modeled as $\mathbf{Y}_{\text{SI}} = \mathbf{H}_{\text{SI}}\mathbf{R}$. Notably, due to the fixed relative positions of the BS transmit and receive antennas, the self-interference channel $\mathbf{H}_{\text{SI}}$ can be treated as deterministic [11].

## III. Problem Formulation and Background on Variational Bayesian Inference

In this section, we begin by formulating the joint radar parameter estimation and uplink communication problem within the Bayesian inference framework. Following this, we provide the background on variational Bayesian inference, which will be utilized in subsequent sections for our proposed algorithms.

### A. Problem Formulation

In the target tracking/uplink communication stage, the mixture of sensing echo, communication signal, and self-interference signal received by the BS can be represented by

$$\tilde{\mathbf{Y}} = \mathbf{Y}_{\text{r}} + \mathbf{Y}_{\text{c}} + \mathbf{Y}_{\text{SI}} = \mathbf{H}_{\text{r}}\mathbf{R} + \mathbf{H}_{\text{c}}\mathbf{X} + \mathbf{H}_{\text{SI}}\mathbf{R} + \mathbf{N}, \tag{9}$$

where $\mathbf{H}_{\text{r}} = \mathbf{A}\left(N_r, \Theta\right) \text{diag}\left(\boldsymbol{\alpha}\right) \mathbf{A}^H\left(N_t, \Theta\right)$ and $\mathbf{N} = \mathbf{N}_{\text{r}} + \mathbf{N}_{\text{c}}$ with $\text{vec}(\mathbf{N}) \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, N_0\mathbf{I}_{N_rS})$. Importantly, a well-designed antenna array can achieve high levels of isolation between the transmit and receive antennas, thus significantly reducing the self-interference power. Moreover, given the deterministic nature of $\mathbf{H}_{\text{SI}}$, it can be accurately estimated as $\hat{\mathbf{H}}_{\text{SI}}$. Typically, the average error power $\frac{1}{N_rN_t}\|\hat{\mathbf{H}}_{\text{SI}} - \mathbf{H}_{\text{SI}}\|^2$ can fall below $-130$ dB [11]. Additionally, as indicated in [23], any remaining self-interference can be further suppressed using radar image processing techniques. Consequently, we assume that self-interference can be effectively mitigated to a negligible level. We redefine the received signal as $\mathbf{Y} = \tilde{\mathbf{Y}} - \mathbf{H}_{\text{SI}}\mathbf{R}$. For the joint target tracking and uplink data detection task, the ISAC BS aims to estimate the radar parameters $\{\Theta, \boldsymbol{\alpha}\}$ and recover the transmitted signal $\mathbf{X}$ at the same time with knowledge of the received signal $\mathbf{Y}$, the radar probing signal $\mathbf{R}$, and the communication CSI $\mathbf{H}_{\text{c}}$.

It is noteworthy that although the radar echo and communication signals interfere with each other in (9), the mutual interference barely degrades the angle estimation performance [10]. Therefore, we can firstly utilize some powerful algorithms to estimate the angle $\Theta$ before decoupling the radar echo and the communication signals.

**Remark 1.** The azimuth angles are usually considered to be constant over multiple channel coherence blocks (quasi-static block-fading channel) [24], [25] or can be assumed to be perfectly predicted [12], [26]. In this case, the angle information only needs to be estimated in the first few channel coherence blocks. Therefore, the main focus of this study is the joint estimation and detection of the reflection coefficients and uplink signals within one coherence block.

Define $\mathbf{y} = \text{vec}(\mathbf{Y})$, $\mathbf{n} = \text{vec}(\mathbf{N})$ and $\mathbf{x} = \text{vec}(\mathbf{X})$. To estimate reflection coefficients and decode $\mathbf{x}$, we rewrite (9) using the equality $\text{vec}(\mathbf{AB}) = (\mathbf{I} \otimes \mathbf{A})\text{vec}(\mathbf{B})$ as

$$\mathbf{y} = \mathbf{G}\boldsymbol{\alpha} + \mathbf{H}\mathbf{x} + \mathbf{n}, \tag{10}$$

where $\mathbf{H} = \mathbf{I} \otimes \mathbf{H}_{\text{c}}$, $\mathbf{G} = [\mathbf{G}_1^T, ..., \mathbf{G}_S^T]^T$, and

$$\mathbf{G}_s = \mathbf{A}(N_r, \Theta)\text{diag}(\mathbf{A}^H(N_t, \Theta)\mathbf{r}[s]). \tag{11}$$

The posterior distribution of $\boldsymbol{\alpha}$ and $\mathbf{x}$ is formulated as

$$p(\mathbf{x}, \boldsymbol{\alpha}|\mathbf{y}; \mathbf{H}_{\text{c}}, \mathbf{G}) \propto p(\mathbf{y}|\mathbf{x}, \boldsymbol{\alpha}; \mathbf{H}_{\text{c}}, \mathbf{G})p(\mathbf{x})p(\boldsymbol{\alpha}), \tag{12}$$

where $p(\mathbf{x}) \propto \prod_{i=1}^{SK} \mathbb{I}_{x_i \in \mathcal{X}}$ and $p(\boldsymbol{\alpha}) \propto \prod_l \mathcal{N}_{\mathbb{C}}(\mu_l, 1)$. Specifically, the mean $\mu_l$ of $\boldsymbol{\alpha}$ is assumed to be acquired in the target searching stage. By removing the term $\mathbf{G}\boldsymbol{\mu}$ from $\mathbf{y}$, we define $\tilde{\boldsymbol{\alpha}} = \boldsymbol{\alpha} - \boldsymbol{\mu} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{I})$ to replace the notation $\boldsymbol{\alpha}$ in (10) and (12), where $\boldsymbol{\mu} = [\mu_1, ..., \mu_L]^T$. Therefore, the Maximum A Posteriori (MAP) solution can be given by

$$\{\mathbf{x}, \tilde{\boldsymbol{\alpha}}\}_{\text{MAP}} = \arg\min_{\mathbf{x}, \tilde{\boldsymbol{\alpha}}} p(\mathbf{x}, \tilde{\boldsymbol{\alpha}}|\mathbf{y}; \mathbf{H}_{\text{c}}, \mathbf{G}). \tag{13}$$

A heuristic linear MMSE (LMMSE) estimation of $\{\mathbf{x}, \tilde{\boldsymbol{\alpha}}\}$ is

$$\begin{bmatrix} \mathbf{x} \\ \tilde{\boldsymbol{\alpha}} \end{bmatrix} = \left(\mathbf{C}^H\mathbf{C} + N_0\mathbf{I}\right)^{-1}\mathbf{C}^H\mathbf{y}, \tag{14a}$$

$$\mathbf{C} = \left[\mathbf{H}, \mathbf{G}\right]. \tag{14b}$$

It is evident that the solution given in (14) achieves global MMSE optimality when all the estimation parameters conform to the standard normal distribution. However, since the transmitted communication signal $\mathbf{x}$ follows a discrete uniform distribution, the estimator in (14) is no longer MMSE-optimal. Besides, the inversion operation of large-dimensional matrix in (14a) could lead to an unacceptable computational burden with the increase of antenna size and coherence time. Therefore, it is imperative to explore more efficient algorithms for the joint radar parameter estimation and communication signal detection problem.

### B. Background on Variational Bayesian Inference

Suppose $\mathbf{x}$ is the signal vector to be estimated, and $\mathbf{y}$ denotes the observed signal vector. The fundamental idea of variational Bayesian inference is to approximate an intractable posterior distribution $p(\mathbf{x}|\mathbf{y})$ using simple distributions $q(\mathbf{x})$ belonging to a predefined family $\mathcal{Q}$. Specifically, one way to achieve this goal is to minimize the Gibbs free energy, defined as [27]

$$\mathcal{F}_{\text{Gibbs}}(q) = U(q) - H(q) = \mathbb{E}_{q(\mathbf{x})}[E(\mathbf{x})] + \mathbb{E}_{q(\mathbf{x})}[\ln q(\mathbf{x})], \tag{15}$$

where $U(q)$ is the variational average energy and $H(q)$ is the variational entropy. The state energy $E(\mathbf{x})$ is given by *Boltzmann's Law*

$$p(\mathbf{x}|\mathbf{y}) = \frac{1}{Z}\exp(-E(\mathbf{x})), \tag{16}$$

where $Z = \int_{-\infty}^{+\infty} \mathrm{d}\mathbf{x} \exp(-E(\mathbf{x}))$ is a normalization constant. In the case of posterior distribution inference problem, $E(\mathbf{x}) = -\ln p(\mathbf{x}, \mathbf{y})$. It is obvious that the Gibbs free energy can be reformulated as

$$\mathcal{F}_{\text{Gibbs}}(q) = \text{KL}[q(\mathbf{x})\|p(\mathbf{x}|\mathbf{y})] - \ln Z, \tag{17}$$

where $\text{KL}[\cdot\|\cdot]$ denotes the KL divergence between two distributions. As $\ln Z$ is a constant, minimizing the Gibbs free energy is equivalent to minimizing the KL divergence.

Since the variational distribution $q(\mathbf{x})$ can take many different forms, there are multiple approaches to minimize the Gibbs free energy. One efficient but simple idea is to sequentially update $q(\mathbf{x})$ by fixing certain components of the distribution.

We assume that $q(\mathbf{x})$ can be factorized as

$$q(\mathbf{x}) = \prod_i q(\mathbf{x}_i), \tag{18}$$

where each trial probability function $q(\mathbf{x}_i)$ is normalized. Assume the distribution $q(\mathbf{x}_{-i}) = \prod_{j\neq i} q(\mathbf{x}_j)$ has already been a good approximation to the distribution $p(\mathbf{x}_{-i}|\mathbf{y})$, where $\mathbf{x}_{-i}$ denotes all elements of $\mathbf{x}$ except $\mathbf{x}_i$. Then, the marginal distribution $q(\mathbf{x}_i)$ can be updated as

$$q(\mathbf{x}_i) \propto \exp\{\mathbb{E}_{q(\mathbf{x}_{-i})}[\ln p(\mathbf{y}|\mathbf{x}) + \ln p(\mathbf{x})]\}. \tag{19}$$

By iteratively computing $q(\mathbf{x}_i)$ as in (19), a mean-field approximation for the belief $p(\mathbf{x}_i|\mathbf{y})$ can be obtained. Here, the Gibbs free energy with $q(\mathbf{x})$ in the form of (18) is called the mean-field free energy.

Another method to minimize the divergence between $p(\mathbf{x}|\mathbf{y})$ and $q(\mathbf{x})$ is to approximate the originally intractable Gibbs free energy using region based methods. Usually, $E(\mathbf{x})$ takes the following form

$$E(\mathbf{x}) = -\sum_a \ln f_a(\mathbf{x}_a), \tag{20}$$

where the argument $\mathbf{x}_a$ of function $f_a$ is a truncated vector of $\mathbf{x}$. One simple version of the region based approximation−the Bethe free energy is defined as [27]

$$\begin{aligned}\mathcal{F}_{\text{Bethe}} = &-\sum_a \mathbb{E}_{q(\mathbf{x}_a)} \ln f_a(\mathbf{x}_a) + \sum_a \mathbb{E}_{q(\mathbf{x}_a)} \ln q(\mathbf{x}_a) \\ &- \sum_i (|d_i| - 1)\mathbb{E}_{q(x_i)} \ln q(x_i),\end{aligned} \tag{21}$$

where $|d_i|$ represents the number of functions $f_a$ that incorporates the argument $x_i$. By minimizing the Bethe free energy, we can obtain efficient message passing algorithms to update the variational distribution $q$.

## IV. Proposed ISAC Variational Bayesian Receiver

In this section, we introduce two variational Bayesian message passing methods for joint radar parameter estimation and communication signal detection. Drawing inspiration from the framework in [28], we begin by employing the mean-field approximation to approximate the exact posterior distribution of $\tilde{\boldsymbol{\alpha}}$ and $\mathbf{x}$, leading to the development of a mean-field method. Additionally, by minimizing the Bethe free energy, we derive a set of fixed-point equations integrated with a noise-variance learning method to further enhance performance.

### *A. The Mean-Field Method*

Following the conventional coordinate ascent variational inference steps, the mean-field method sequentially updates the coefficients $\tilde{\boldsymbol{\alpha}}$ and the communication signals $\mathbf{x}$ as outlined in (19). The variational distribution to approximate the posterior probability of $\tilde{\boldsymbol{\alpha}}$ and $\mathbf{x}$ is expressed as

$$q(\mathbf{x}, \tilde{\boldsymbol{\alpha}}) = \left[\prod_s^S \prod_i^K q(x_i[s])\right] q(\tilde{\boldsymbol{\alpha}}), \tag{22}$$

where $\mathbf{x} = [\mathbf{x}^T[1], ..., \mathbf{x}^T[S]]^T \in \mathbb{C}^{SK}$, and $x_i[s]$ denotes the $i$-th element of vector $\mathbf{x}[s] \in \mathbb{C}^K$. Notably, the sensing parameters are modeled as a vector to capture the posterior correlations among the sensing coefficients, while the more numerous communication signals are fully factorized to facilitate lower-complexity processing. The mean of the marginal distribution of $q(\mathbf{x}, \tilde{\boldsymbol{\alpha}})$ serves as the inference for $\tilde{\alpha}$ and $\mathbf{x}$.

**Algorithm 1:** The Mean-Field Method

**Input :** $\mathbf{Y}, \mathbf{H}_\text{c}, \mathbf{G}, p(\mathbf{x}), p(\tilde{\boldsymbol{\alpha}}), S, T.$
**Output:** $\hat{\mathbf{x}}_\text{c}, \hat{\tilde{\boldsymbol{\alpha}}}.$

- **Initialization:** $\hat{x}_i = 0, \sigma^2_{\hat{x}_i} = 1, \hat{\tilde{\boldsymbol{\alpha}}} = \mathbf{0}, \delta[s] = N_0, p(\mathbf{x}) \propto \prod_{i=1}^{SK} \mathbb{I}_{x_i \in \mathcal{X}}, p(\tilde{\boldsymbol{\alpha}}) \propto \prod_l \mathcal{N}_\mathbb{C}(0, 1).$
- **for** $t = 1, 2, ..., T$ **do**
  - $\mathbb{E}_{q^{\setminus \tilde{\alpha}}}[\mathbf{X}_\text{c}] = \hat{\mathbf{X}}.$
  - **Update reflection coefficients** $\tilde{\boldsymbol{\alpha}}$:
  - Calculate the variational mean $\hat{\tilde{\boldsymbol{\alpha}}}$ and variance $\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}$ of $\tilde{\boldsymbol{\alpha}}$ as in (24) and (25) with noise variance $N_0$ replaced by $\langle \delta[s] \rangle_s$.
  - $\mathbb{E}_{q^{\setminus i}}[\tilde{\boldsymbol{\alpha}}] = \hat{\tilde{\boldsymbol{\alpha}}}.$
  - **Update communication signals** $\mathbf{x}$:
  - **for** $s = 1, 2, ..., S$ **do**
    - **for** $i = 1, 2, ..., K$ **do**
      - Compute $z_i[s]$ as in (28),
      - $\xi_i[s] = \delta[s]/\|\mathbf{h}_i\|^2.$
      - Calculate the variational mean $\hat{x}_i[s]$ and variance $\sigma^2_{\hat{x}_i[s]}$ of $x_i[s]$ as in (29).
  - **Update noise variance** $\delta[s], s = 1, 2, ..., S$:
  - Calculate $\delta[s]$ as in (33).

1) **Update reflection coefficients** $\tilde{\boldsymbol{\alpha}}$: The optimization of $q(\mathbf{x}, \tilde{\boldsymbol{\alpha}})$ starts with the update of $q(\tilde{\boldsymbol{\alpha}})$. Denoting $\prod_s \prod_i q(x_i[s])$ as $q^{\setminus \tilde{\boldsymbol{\alpha}}}$, we compute $q(\tilde{\boldsymbol{\alpha}})$ as follows

$$\begin{aligned} q(\tilde{\boldsymbol{\alpha}}) &\propto \exp\{\mathbb{E}_{q^{\setminus \tilde{\alpha}}}[\ln p(\mathbf{y}|\mathbf{x}, \tilde{\boldsymbol{\alpha}}; \mathbf{H}_\text{c}, \mathbf{G}) + \ln p(\tilde{\boldsymbol{\alpha}})]\} \\ &\propto \exp\{\ln p(\tilde{\boldsymbol{\alpha}}) - \mathbb{E}_{q^{\setminus \tilde{\alpha}}}[\frac{1}{N_0}\|\mathbf{y} - \mathbf{H}\mathbf{x} - \mathbf{G}\tilde{\boldsymbol{\alpha}}\|^2]\} \\ &\propto \exp\{-\frac{1}{N_0}\|\mathbf{y} - \mathbf{H}\mathbb{E}_{q^{\setminus \tilde{\alpha}}}[\mathbf{x}] - \mathbf{G}\tilde{\boldsymbol{\alpha}}\|^2 - \tilde{\boldsymbol{\alpha}}^H\tilde{\boldsymbol{\alpha}}\}. \end{aligned} \tag{23}$$

Taking the expectation with respect to the variational distribution $q(\tilde{\boldsymbol{\alpha}})$, the estimate of $\tilde{\boldsymbol{\alpha}}$ is obtained by

$$\hat{\tilde{\boldsymbol{\alpha}}} = (\mathbf{G}^H\mathbf{G} + N_0\mathbf{I})^{-1}\mathbf{G}^H(\mathbf{y} - \mathbf{H}\mathbb{E}_{q^{\setminus \tilde{\alpha}}}[\mathbf{x}]), \tag{24}$$

which corresponds to the linear MMSE estimator. The covariance matrix of $\hat{\tilde{\boldsymbol{\alpha}}}$ is given as

$$\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}} = (\mathbf{G}^H\mathbf{G}/N_0 + \mathbf{I})^{-1}. \tag{25}$$

2) **Update communication signals** $\mathbf{x}$: The next step involves the sequential update of $q(x_i[s])$. Before that, we first restate an important property on the variational posterior mean as elucidated in [29].

**Property 1.** Let $\mathbf{x} \in \mathbb{C}^m$ denote a random vector, which follows the variational distribution $q(\mathbf{x})$. Suppose an arbitrary matrix $\mathbf{A} \in \mathbb{C}^{n\times m}$ and a vector $\mathbf{y} \in \mathbb{C}^n$ are both deterministic. Let $\boldsymbol{\Sigma}_\mathbf{x}$ denote covariance matrix of $\mathbf{x}$. Then, the expectation of $\|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2$ with respect to $q(\mathbf{x})$ is given by

$$\mathbb{E}_{q(\mathbf{x})}[\|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2] = \|\mathbf{y} - \mathbf{A}\mathbb{E}_{q(\mathbf{x})}[\mathbf{x}]\|^2 + \text{Tr}\{\mathbf{A}\boldsymbol{\Sigma}_\mathbf{x}\mathbf{A}^H\}. \tag{26}$$

$$\mathcal{F}_{\mathrm{MF}}(q) = \mathrm{KL}(q\|p_0) + \sum_{s=1}^{S}[N_r \ln(\pi\delta[s]) + \frac{1}{\delta[s]}(\|\mathbf{y}[s] - \mathbf{H}_{\mathrm{c}}\hat{\mathbf{x}}[s] - \mathbf{G}_s\hat{\tilde{\boldsymbol{\alpha}}}\|^2 + \mathrm{Tr}\{\mathbf{H}_{\mathrm{c}}\boldsymbol{\Sigma}_{\hat{\mathbf{x}}[s]}\mathbf{H}_{\mathrm{c}}^H\} + \mathrm{Tr}\{\mathbf{G}_s\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}\mathbf{G}_s^H\})]. \tag{30}$$

$$\begin{aligned}\mathrm{KL}(q\|p_0) &= \sum_{s=1}^{S}\mathbb{E}_{q(\mathbf{x}[s],\tilde{\boldsymbol{\alpha}})}[\ln\frac{q(\mathbf{x}[s],\tilde{\boldsymbol{\alpha}})}{p(\mathbf{x}[s])p(\tilde{\boldsymbol{\alpha}})}] = -\sum_{s=1}^{S}\Big\{\sum_{i=1}^{K}[\ln Z(z_i[s],\xi_i[s]) + \frac{\sigma^2_{\hat{x}_i[s]} + |\hat{x}_i[s] - z_i[s]|^2}{\xi_i[s]}] \\ &\quad - \mathrm{Tr}\{\boldsymbol{\Sigma}_{\tilde{\boldsymbol{\alpha}}}^{-1}[s][\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}} + (\hat{\tilde{\boldsymbol{\alpha}}} - \boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}[s])(\hat{\tilde{\boldsymbol{\alpha}}}^H - \boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}^H[s])]\}\Big\} - \ln Z(\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}, \boldsymbol{\Sigma}_{\tilde{\boldsymbol{\alpha}}}).\end{aligned} \tag{31}$$

$$\begin{aligned}\mathcal{F}_{\mathrm{Bethe}}(q) &= \mathrm{KL}^*(q\|p_0) + \sum_{s=1}^{S}\Big\{N_r\ln(\pi\delta[s]) + \frac{1}{\delta[s]}[\|\mathbf{y}[s] - \mathbf{H}_{\mathrm{c}}\hat{\mathbf{x}}[s] - \mathbf{G}_s\hat{\tilde{\boldsymbol{\alpha}}}\|^2] \\ &\quad + \sum_{n=1}^{N_r}\ln[1 + \sum_{i=1}^{K}(|h_{ni}|^2\sigma^2_{\hat{x}_i[s]})/\delta[s] + \mathbf{g}_n^H[s]\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}\mathbf{g}_n[s]/\delta[s]]\Big\}.\end{aligned} \tag{32}$$

*Proof.* The proof of Property 1 can be found in [30]. ∎

Let $q^{\setminus i[s]}$ represent $\prod_r \prod_j q(x_j[r])q(\tilde{\boldsymbol{\alpha}})/q(x_i[s])$. According to Property 1, the marginal distribution $q(x_i[s])$, $i = 1, 2, ..., K$, can be computed as

$$\begin{aligned}q(x_i[s]) &\propto \exp\{\mathbb{E}_{q^{\setminus i[s]}}[\ln p(\mathbf{y}|\mathbf{x}, \tilde{\boldsymbol{\alpha}}; \mathbf{H}_{\mathrm{c}}, \mathbf{G}) + \ln p(\mathbf{x})]\}, \\ &\propto \exp\{\ln p(\mathbf{x}) - \mathbb{E}_{q^{\setminus i[s]}}[\frac{1}{N_0}\|\mathbf{y} - \mathbf{H}\mathbf{x} - \mathbf{G}\tilde{\boldsymbol{\alpha}}\|^2]\}, \\ &\propto p(x_i[s])\exp\{-\frac{1}{N_0}\|\mathbf{h}_i\|^2|x_i[s] - z_i[s]|^2\},\end{aligned} \tag{27}$$

where $\mathbf{h}_i$ denotes the $i$-th column of matrix $\mathbf{H}_{\mathrm{c}}$, and $z_i[s]$ is then calculated as

$$z_i[s] = \frac{\mathbf{h}_i^H}{\|\mathbf{h}_i\|^2}(\mathbf{y}[s] - \sum_{j\neq i}\mathbf{h}_j\mathbb{E}_{q^{\setminus i[s]}}[x_j[s]] - \mathbf{G}_s\mathbb{E}_{q^{\setminus i[s]}}[\tilde{\boldsymbol{\alpha}}]). \tag{28}$$

Let $N_0/\|\mathbf{h}_i\|^2 = \xi_i$. The mean and variance of $q(x_i[s])$ are given by

$$\hat{x}_i[s] = F_{\mathrm{a}}(z_i[s], \xi_i), \sigma^2_{\hat{x}_i[s]} = F_{\mathrm{c}}(z_i[s], \xi_i), \tag{29}$$

where $\hat{\mathbf{x}} = [\hat{\mathbf{x}}^T[1], ..., \hat{\mathbf{x}}^T[S]]^T$ with $\hat{\mathbf{x}}[s] = [\hat{x}_1[s], ..., \hat{x}_K[s]]^T, s = 1, 2, ..., S$ can be interpreted as the posterior estimation of $\mathbf{x}$.

3) **Update noise variance** $\delta[s]$ : Moreover, it has been demonstrated that incorporating noise variance learning into the framework of the mean-field method can remarkably enhance performance [28]. In other words, the noise variance $N_0$ is treated as an unknown variable to be estimated, denoted as $\delta$. Instead of specifying a prior distribution to the noise variance as in [29], a direct minimization of mean-field free energy [28] is applied here to update noise variance in each iteration. We denote $\delta[s]$ as the noise variance in the $s$-th snapshot, and then present the following theorem.

**Theorem 1.** The formulation of mean-field free energy of the system model can be given by (30) and (31).

*Proof.* Theorem 1 is proven in Appendix A. ∎

By setting the derivatives of the mean-field free energy with respect to each $\delta[s]$ to zero, we obtain the fixed-point equations for the noise variance

$$\delta[s] = \frac{a_0[s] + b_0[s]}{N_r}, \tag{33}$$

where $a_0[s] = \mathrm{Tr}\{\mathbf{H}_{\mathrm{c}}\boldsymbol{\Sigma}_{\hat{\mathbf{x}}[s]}\mathbf{H}_{\mathrm{c}}^H\} + \mathrm{Tr}\{\mathbf{G}_s\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}\mathbf{G}_s^H\}$ characterizes the equivalent noise power induced by posterior estimation uncertainty of $\hat{\mathbf{x}}[s]$ and $\hat{\tilde{\boldsymbol{\alpha}}}$, and $b_0[s] = \|\mathbf{y}[s] - \mathbf{H}_{\mathrm{c}}\hat{\mathbf{x}}[s] - \mathbf{G}_s\hat{\tilde{\boldsymbol{\alpha}}}\|^2$ measures both the true physical noise variance and error power due to inaccurate estimates of $\mathbf{x}[s]$ and $\tilde{\boldsymbol{\alpha}}$. Given that the diagonal terms of $\mathbf{G}^H\mathbf{G}$ are significantly dominant compared to the off-diagonal elements, the covariance matrix in (33) can be approximated as a diagonal matrix as $\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}} = \mathrm{diag}([\sigma_{\hat{\tilde{\alpha}}_1}, ..., \sigma_{\hat{\tilde{\alpha}}_L}]^T)$.

The mean-field method, as summarized in Alg. 1, operates in an iterative manner, where $t = 1, 2, ..., T$ signifies the iteration index. It is noteworthy that $\delta[s]$ can be directly initialized to a constant. Meanwhile, $\xi_i$ is updated to $\xi_i[s] = \delta[s]/\|\mathbf{h}_i\|^2$. Additionally, the noise variance $N_0$ in (24) and (25) can be replaced by the arithmetic mean of $\delta[s]$ as $\langle\delta[s]\rangle_s = (\sum_s \delta[s])/S$ to reduce complexity.

### B. The Bethe Method

Although the mean-field method offers a relatively accessible set of fixed-point equations for approximating the exact posterior distribution, the factorized form of the mean-field approximation in (22) relies on a heuristic assumption of independence among variables, which limits its ability to model the conditional dependencies among the elements of $\{\mathbf{x}, \tilde{\boldsymbol{\alpha}}\}$ given $\mathbf{y}$. Moreover, the mean-field method employs a fully sequential update framework, which may hinder the implementation efficiency of the algorithm. While this approach simplifies the derivation, it may introduce significant processing delays as the size of the ISAC system increases.

Instead of considering other complicated form for $q(\mathbf{x}, \tilde{\boldsymbol{\alpha}})$, the Bethe free energy based method provides a region-based approximation to the Gibbs free energy by considering local correlations among variables [31]. By preserving the structural information of the probabilistic model, the Bethe framework offers significantly more accurate approximations of the exact free energy than the mean-field method. Furthermore, the proposed Bethe method builds upon the architecture of the mean-field method, employing a similar update schedule for the reflection parameters, communication symbols, and noise variance.

Let $h_{ni}$ denote the element in the $n$-th row and $i$-th column of $\mathbf{H}_{\rm c}$, and let $\mathbf{g}_n[s]$ denote the $n$-th row of $\mathbf{G}_s$. Then, the following theorem is established by utilizing the fixed-point properties of belief propagation (BP).

**Theorem 2.** The formulation of Bethe free energy of the system model is given by (32).

*Proof.* Theorem 2 is proven in Appendix B. ∎

For jointly estimating $\tilde{\boldsymbol{\alpha}}$ and detecting $\mathbf{x}$, a set of variational message passing fixed-point equations can be obtained by investigating the stationary points of unconstrained Bethe free energy as in (66). Under the assumption of unconstrained conditions, the extrinsic mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\Sigma}_{\boldsymbol{\mu}}$ are assumed to be independent from $\hat{\tilde{\boldsymbol{\alpha}}}$ and $\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}$. Similarly, $z_i[s]$ and $\xi_i[s]$ are assumed to be independent of $\hat{x}_i[s]$ and $\sigma^2_{\hat{x}_i[s]}$. For clarity, we denote the iteration number in the update process of each message with the superscript $t$. In the $t+1$-th round of iteration, the intermediate parameter $V_n^{t+1}[s]$ is firstly calculated as in (70), and $\omega_n^{t+1}[s]$ is obtained by

$$\omega_n^{t+1}[s] = \sum_{i=1}^{K} h_{ni}\hat{x}_i[s] + \mathbf{g}_n[s]\hat{\tilde{\boldsymbol{\alpha}}} - \frac{(y_n[s]-\omega_n^t[s])V_n^{t+1}[s]}{\delta[s]+V_n^t[s]}. \tag{34}$$

1) **Update reflection coefficients** $\tilde{\boldsymbol{\alpha}}$: By determining the zero points of the derivative of (66) with respect to $\boldsymbol{\Sigma}_{\boldsymbol{\mu}}$, $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}$ and $\hat{\tilde{\boldsymbol{\alpha}}}$, the iterative equations for estimating $\tilde{\boldsymbol{\alpha}}$ are specified as follows

$$\boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{t+1} = \left(\sum_{s=1}^{S}\sum_{n=1}^{N_r}\frac{\mathbf{g}_n^H[s]\mathbf{g}_n[s]}{\delta[s]+V_n^{t+1}[s]}\right)^{-1}, \tag{35}$$

$$\boldsymbol{\mu}^{t+1} = \hat{\tilde{\boldsymbol{\alpha}}}^t + \boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{t+1}\sum_{s=1}^{S}\sum_{n=1}^{N_r}\mathbf{g}_n^*[s]\frac{y_n[s]-\omega_n^{t+1}[s]}{\delta[s]+V_n^{t+1}[s]}, \tag{36}$$

$$\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}^{t+1} = \left((\boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{t+1})^{-1}+\mathbf{I}\right)^{-1}, \tag{37}$$

$$\hat{\tilde{\boldsymbol{\alpha}}}^{t+1} = \left(\boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{t+1}+\mathbf{I}\right)^{-1}\boldsymbol{\mu}^{t+1}. \tag{38}$$

In (37) and (38), multiple matrix inversion operations are involved, which can significantly increase computational complexity. To mitigate this, we can directly solve the fixed-point formula of $\hat{\tilde{\boldsymbol{\alpha}}}^{t+1}$ from (38). Specifically, we first substitute the fixed-point (71) into (36), and then incorporate the resulting expression into (38). This process reveals that the solution for $\hat{\tilde{\boldsymbol{\alpha}}}^{t+1}$ exactly corresponds to the expression given in (24).

2) **Update communication signals** $\mathbf{x}$: As for detecting $\mathbf{x}$, the iterative equations can be obtained by setting the partial derivative of (66) with respect to $\xi_i[s]$, $z_i[s]$, $\hat{x}_i[s]$ and $\sigma^2_{\hat{x}_i[s]}$ to zero. This results in a set of general approximate message passing (GAMP) formulas as in [32], which are given by

$$\xi_i^{t+1}[s] = \left(\sum_{n=1}^{N_r}\frac{|h_{ni}|^2}{\delta[s]+V_n^{t+1}[s]}\right)^{-1}, \tag{39}$$

$$z_i^{t+1}[s] = \hat{x}_i^t[s] + \xi_i^{t+1}[s]\sum_{n=1}^{N_r}h_{ni}^*\frac{y_n[s]-\omega_n^{t+1}[s]}{\delta[s]+V_n^{t+1}[s]}, \tag{40}$$

**Algorithm 2:** The Bethe Method

**Input :** $\mathbf{Y}, \mathbf{H}_{\rm c}, \mathbf{G}, p(\mathbf{x}), p(\tilde{\boldsymbol{\alpha}}), S, T$.
**Output:** $\hat{\mathbf{x}}_{\rm c}, \hat{\tilde{\boldsymbol{\alpha}}}$.

**Initialization:** $\hat{x}_i = 0, \sigma^2_{\hat{x}_i} = 1, \hat{\tilde{\boldsymbol{\alpha}}} = \mathbf{0}, \delta[s] = N_0, p(\mathbf{x}) \propto \prod_{i=1}^{SK}\mathbb{I}_{x_i\in\mathcal{X}}, p(\tilde{\boldsymbol{\alpha}}) \propto \prod_l \mathcal{N}_{\mathbb{C}}(0,1)$.

**for** $t = 1,2,...,T$ **do**

- **Update reflection coefficients** $\tilde{\boldsymbol{\alpha}}$:
- Calculate the variational mean $\hat{\tilde{\boldsymbol{\alpha}}}$ and variance $\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}$ of $\tilde{\boldsymbol{\alpha}}$ as in (24) and (37).
- **Update communication signals** $\mathbf{x}$:
- **for** $s = 1,2,...,S$ **do**
  - Compute $V_n^{t+1}[s]$ and $\omega_n^{t+1}[s]$ as in (70) and (41) for each $n = 1,2,...,N_r$, parallelly.
  - Compute $\xi_i^{t+1}[s]$ and $z_i^{t+1}[s]$ as in (39) and (40) for each $i = 1,2,...,K$, parallelly.
- Parallelly calculate the variational mean $\hat{x}_i[s]$ and variance $\sigma^2_{\hat{x}_i[s]}$ of $x_i[s]$ as in (29) for $i = 1,2,...,K$, $s = 1,2,...,S$.
- **Update noise variance** $\delta[s], s = 1,2,...,S$ :
- Calculate $\delta[s]$ as (44).

with $\hat{x}_i[s]$ and $\sigma^2_{\hat{x}_i[s]}$ computed as in (29).

In practice, the power of the user communication signal is significantly smaller than that of the radar echo received by the BS, primarily due to the power constraints of the user terminal. Consequently, simultaneously updating $\mathbf{x}$ and $\tilde{\boldsymbol{\alpha}}$ may significantly degrade the performance of communication detection. Therefore, we first estimate the reflection coefficient $\tilde{\boldsymbol{\alpha}}$ in (24) and (37). Subsequently, similar to the alternating optimization method, in each iteration, we can first remove the radar echo from the received signals. Then, (34) is reformulated as

$$\omega_n^{t+1}[s] = \sum_{i=1}^{K} h_{ni}\hat{x}_i[s] - \frac{(y_n^{\rm c}[s]-\omega_n^t[s])V_n^{t+1}[s]}{\delta[s]+V_n^t[s]}, \tag{41}$$

where $y_n^{\rm c}[s] = y_n[s] - \mathbf{g}_n[s]\hat{\tilde{\boldsymbol{\alpha}}}$. Meanwhile, $y_n[s]$ in (40) is replaced by $y_n^{\rm c}[s]$. By solving $\omega_n$ in (41), the fixed-point solution of $z_i^{t+1}[s]$ can be obtained as

$$z_i^{t+1}[s] = \hat{x}_i^t[s] + \frac{\xi_i^{t+1}[s]}{\delta[s]}\sum_{n=1}^{N_r}h_{ni}^*\Big(y_n^{\rm c}[s] - \sum_{i=1}^{K}h_{ni}\hat{x}_i[s]\Big), \tag{42}$$

which is also consistent with the stationary point of (32).

3) **Update noise variance** $\delta[s]$ : Similar to the mean-field method, the Bethe framework also allows for the updating of the noise variance. We first bring the fixed-point solutions (70) and (71) into the unconstrained free energy to obtain (32). Subsequently, we take the stationary point of (32) with respect to the noise variance, resulting in the following equation

$$\sum_{n=1}^{N_r}\frac{V_n[s]\delta[s]}{\delta[s]+V_n[s]} - N_r\delta[s] + b_0[s] \equiv 0. \tag{43}$$

By assuming $V_n[s]$ is invariant to different indices $n$, an

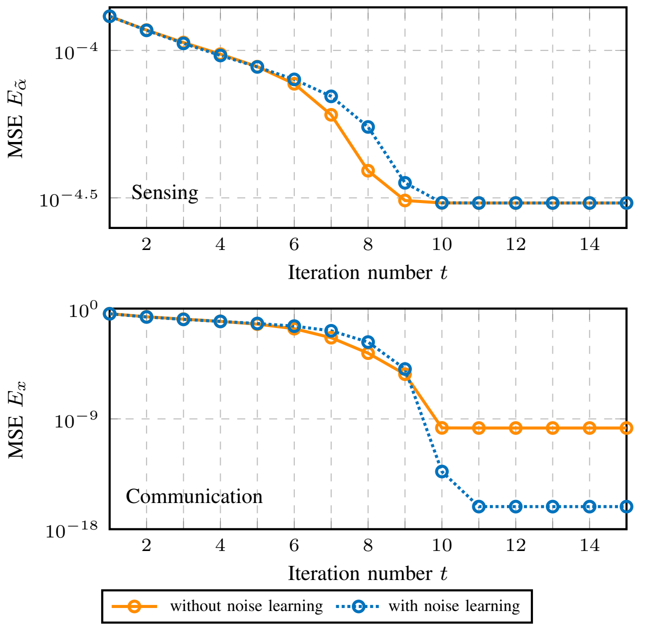



Fig. 2. MSEs performance of sensing and communication under 16-QAM with $N_r = 40$, $K = 30$, and $L = 8$.

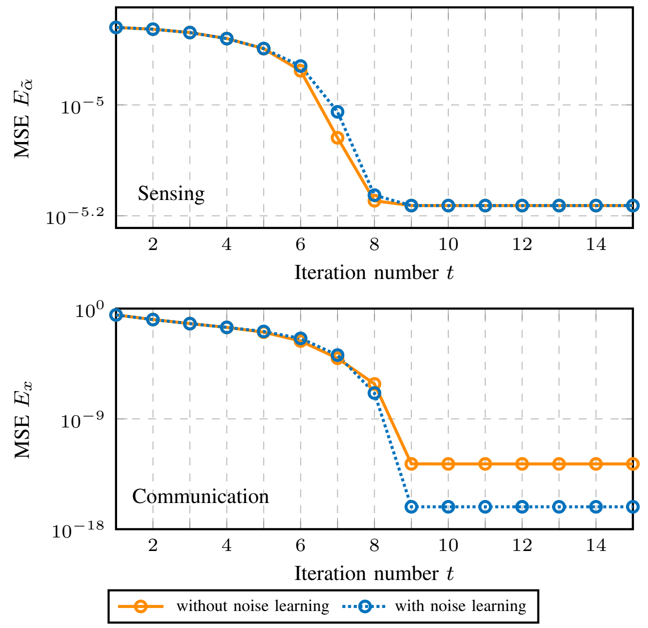



Fig. 3. MSEs performance of sensing and communication under 64-QAM with $N_r = 40$, $K = 20$, and $L = 8$.

approximation of $\delta[s]$ can be obtained by

$$\delta[s] \approx \frac{b_0[s] + [b_0[s]^2 + 4b_0[s](\sum_{n=1}^{N_r} V_n[s])]^{\frac{1}{2}}}{2N_r}. \tag{44}$$

It is not hard to observe that the approximation in (44) is upper-bounded by (33).

**Remark 2.** In recent studies on VI algorithms [20], [33], [34], noise variance is assumed to follow a predefined distribution. And it is modeled as a variable node in the factor graph and then updated using MAP or MMSE estimation criteria. However, the primary goal of VI algorithms is not to precisely estimate the noise variance itself but to enable accurate recovery of the target signals. Motivated by this, the proposed methods treat the noise variance as a hyperparameter, without relying on any hyperprior distribution. Interestingly, the learning of the noise variance shares a conceptual similarity with the M-step of the expectation-maximization (EM) algorithm [35] for modeling distribution parameters.

The Bethe method is summarized in Alg. 2. For further complexity reduction, $V_n[s]$ and the estimated noise variance $\delta[s]$ can be replaced by their arithmetic means in the computations of variance-like parameters (i.e., $\mathbf{\Sigma}_{\boldsymbol{\mu}}$, $\xi_i[s]$). Specifically, $V_n[s]$, and $\delta[s]$ can be approximated as $\langle V_n[s]\rangle_n$, and $\langle \delta[s]\rangle_s$ in (35) and (39), respectively.

## V. Performance Analysis

In this section, we analyze the statistical performance of the proposed Bethe method under asymptotic assumptions. A set of SE equations are derived to evaluate the MSE performance of both sensing and communication. Various approximations are employed to obtain closed-form expressions for the recursive evolution equations. It is important to note that the SE analysis is not applicable to the mean-field method due to its sequential update structure and the resulting statistical dependencies.

In the context of the large system limit, we assume that $N_r$ and $K$ tend to infinity while maintaining a fixed ratio $\kappa = N_r/K$. The number of snapshots $S$ can take any finite value. Furthermore, each entry of channel matrix $\mathbf{H}_c$ is approximated as i.i.d. Gaussian random variables with mean $0$ and variance $1/K$ [36].

For the $t$-th iteration, the SE parameters are defined as follows

$$V_{\tilde{\alpha}}^t = \lim_{L\to\infty} \frac{1}{L}\sum_{l=1}^{L} \sigma_{\hat{\tilde{\alpha}}_l}^{2,t} \overset{\text{a.e.}}{=} \mathbb{E}[\sigma_{\hat{\tilde{\alpha}}_l}^{2,t}], \tag{45}$$

$$E_{\tilde{\alpha}}^t = \lim_{L\to\infty} \frac{1}{L}\sum_{l=1}^{L} |\hat{\tilde{\alpha}}_l^t - \tilde{\alpha}_l|^2 \overset{\text{a.e.}}{=} \mathbb{E}[|\hat{\tilde{\alpha}}_l^t - \tilde{\alpha}_l|^2], \tag{46}$$

$$V_x^t = \lim_{K\to\infty} \frac{1}{K}\sum_{i=1}^{K} \sigma_{\hat{x}_i}^{2,t} \overset{\text{a.e.}}{=} \mathbb{E}[\sigma_{\hat{x}_i}^{2,t}], \tag{47}$$

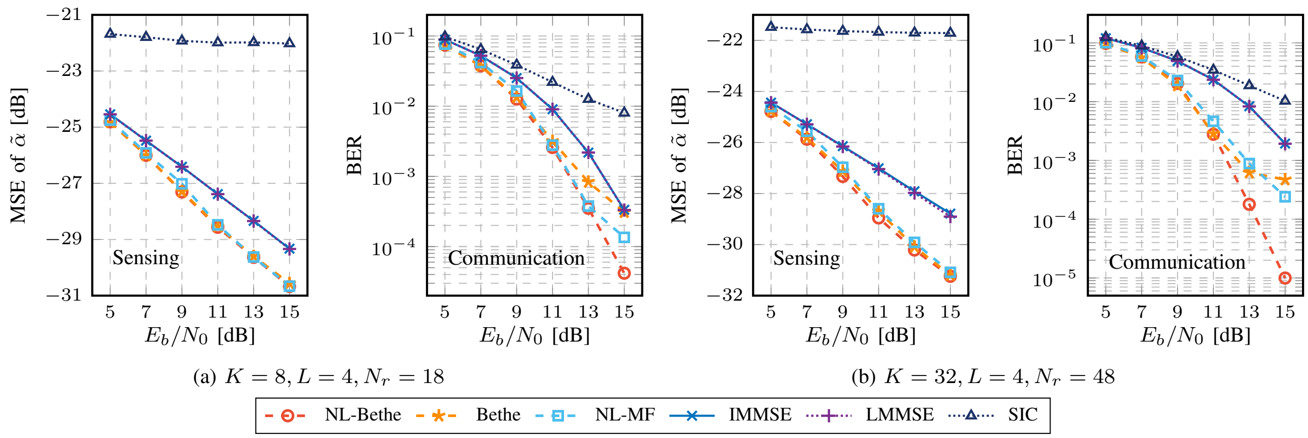
$$E_x^t = \lim_{K\to\infty} \frac{1}{K}\sum_{i=1}^{K} |\hat{x}_i^t - x_i|^2 \overset{\text{a.e.}}{=} \mathbb{E}[|\hat{x}_i^t - x_i|^2], \tag{48}$$

where $V_{\tilde{\alpha}}^t$, $V_x^t$ are the average variance of variational distributions, and $E_{\tilde{\alpha}}^t$, $E_x^t$ represent the average MSE of the estimation $\hat{\tilde{\alpha}}^t$ and $\hat{x}^t$. With a slight abuse of notations, the estimated variational mean and variance $\sigma_{\hat{\tilde{\alpha}}_l}^{2,t}$, $\hat{\tilde{\alpha}}_l^t$, $\sigma_{\hat{x}_i}^{2,t}$, and $\hat{x}_i^t$ are treated as random variables in this discussion. Note that, without loss of generality, we have omitted all indices related to the snapshots $s$ of various parameters for simplicity.

Following the procedure outlined in Alg. 2, $V_{\tilde{\alpha}}^t$ is given by

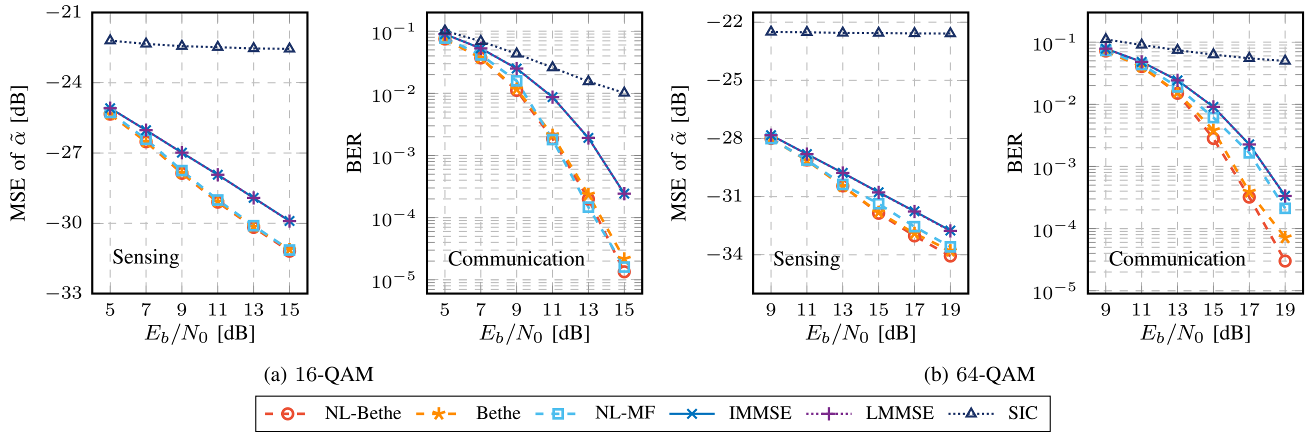
$$V_{\tilde{\alpha}}^t = \frac{1}{L}\mathrm{Tr}(\mathbf{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}^t). \tag{49}$$

According to (24), the MMSE of $\tilde{\alpha}_l$ in each iteration can be




(a) $K = 8, L = 4, N_r = 18$ (b) $K = 32, L = 4, N_r = 48$

Fig. 4. MSE and BER performance of various methods in correlated channels under 16-QAM with $\zeta_R = 0.05$, $\zeta_T = 0.05$, $N_t = 12$, and $S = 10$.




(a) 16-QAM (b) 64-QAM

Fig. 5. MSE and BER performance of various methods in correlated channels under different modulations with $K = 12, L = 8, N_r = 28$, $N_t = 12$, $S = 10$, $\zeta_R = 0.1$ and $\zeta_T = 0.1$.

expressed by

$$E_{\tilde{\alpha}}^{t+1} = \frac{1}{L}\mathrm{Tr}\{\mathbf{I} - \mathbf{G}^H(\mathbf{H}\boldsymbol{\Sigma}_x^t\mathbf{H}^H + \mathbf{G}\mathbf{G}^H + N_0\mathbf{I})^{-1}\mathbf{G}\}, \quad (50)$$

where $\boldsymbol{\Sigma}_x^t = E_x^t\mathbf{I}$. To track the dynamics of $V_x^t$ and $E_x^t$, the variational posterior distribution $q(x_i)$ at the $t+1$-th iteration can be firstly modeled as

$$q^{t+1}(x_i) \propto \mathbb{I}_{x_i \in \mathcal{X}} \cdot \exp\{-\frac{|x_i - z_i^t|^2}{(\delta^t + V_{\tilde{\alpha}}^{\prime t+1} + V_x^t)/\kappa}\}, \quad (51)$$

where $V_{\tilde{\alpha}}^{\prime t+1} = \sum_{l=1}^{L}|g_{al}|^2 V_{\tilde{\alpha}}^t \approx \frac{1}{N_r}\sum_{a,l}|g_{al}|^2 V_{\tilde{\alpha}}^t$, and $\delta^t$ is the noise variance to be learned. Following the same approximations in [37] by neglecting $O(\frac{1}{K})$ terms, $z_i^t$ can be viewed as the sum of two random variables, i.e., $x \sim \mathcal{U}(\mathcal{X})$ and $s \sim \mathcal{N}_{\mathbb{C}}(0, \frac{E_x^t + E_{\tilde{\alpha}}^{\prime t+1} + N_0}{\kappa})$, where $E_{\tilde{\alpha}}^{\prime t+1} = \frac{1}{N_r}\sum_{a,l}|g_{al}|^2 E_{\tilde{\alpha}}^{t+1}$. Given the above results, the evolution equations of $V_x^t$ and $E_x^t$ are calculated as

$$V_x^{t+1} = \mathbb{E}[F_{\mathrm{c}}(x + s, \frac{V_x^t + V_{\tilde{\alpha}}^{\prime t+1} + \delta^t}{\kappa})], \quad (52)$$

$$E_x^{t+1} = \mathbb{E}[|F_{\mathrm{a}}(x + s, \frac{V_x^t + V_{\tilde{\alpha}}^{\prime t+1} + \delta^t}{\kappa}) - x|^2], \quad (53)$$

where the expectation operations $\mathbb{E}[\cdot]$ are with respect to random variables $x$ and $s$. Finally, by taking the expectation on both sides of (43), SE formula for noise variance can be derived as follows

$$\delta^{t+1} = \frac{e + \left[e^2 + 4e(V_{\tilde{\alpha}}^{\prime t+1} + V_x^{t+1})\right]^{\frac{1}{2}}}{2}, \quad (54)$$

where $e = (N_0 + E_{\tilde{\alpha}}^{\prime t+1} + \kappa E_x^{t+1})/N_r$.

It is clear that the SE equations (49-54) describe the convergence performance of Bethe method. However, the measurement matrix element $g_{al}$ in $V_{\tilde{\alpha}}^{\prime t}$ and $E_{\tilde{\alpha}}^{\prime t}$ remains a random variable whose specific distribution is difficult to determine. Moreover, (52) and (53) involve integral operations that do not yield closed-form solutions. Consequently, we employ the Monte Carlo method with respect to $g_{al}$ to simulate the evolution of MSEs $E_{\tilde{\alpha}}^t$ and $E_x^t$.

For the theoretical analysis of the convergence performance, we consider an idealized ISAC system model where both the average power of the communication signals and the variance of the fading coefficients are set to unity. The measurement matrix element is normalized by dividing $g_{al}$ by $\sqrt{K}$. Fig. 2 and Fig. 3 depict the evolution of MSEs for sensing and communication as functions of the iteration number $t$ under 16-QAM and 64-QAM, respectively. Notably, although the SE equations are derived under the large-system limit assumptions, the simulations conducted with a relatively small

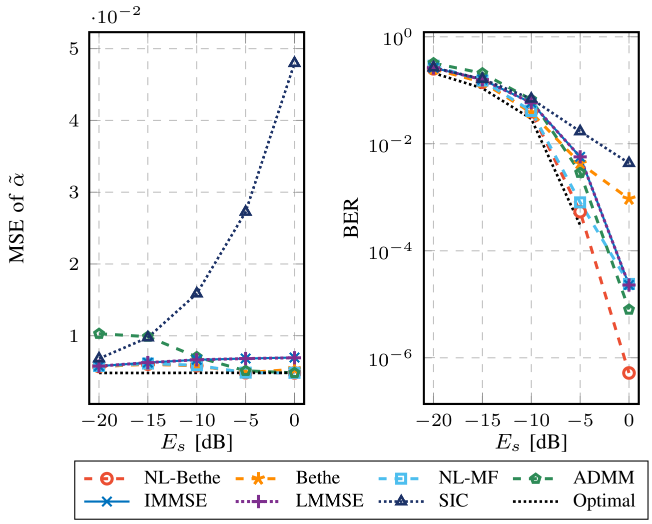



Fig. 6. MSE and BER performance of various methods with respect to the communication signal power $E_s$.

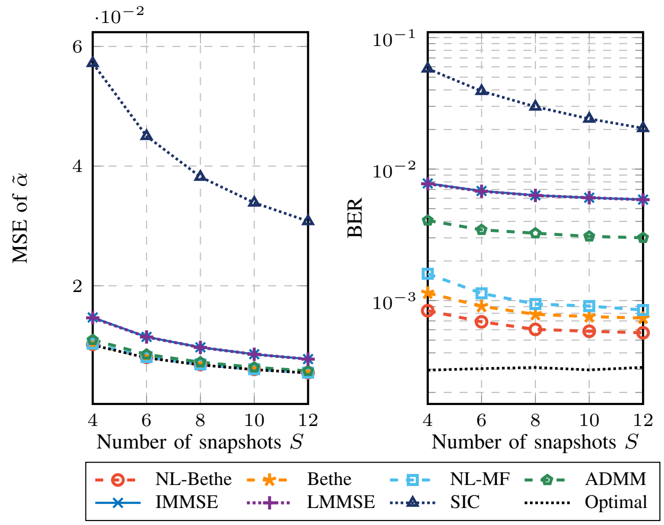



Fig. 8. MSE and BER performance of various methods with respect to the number of snapshots $S$.

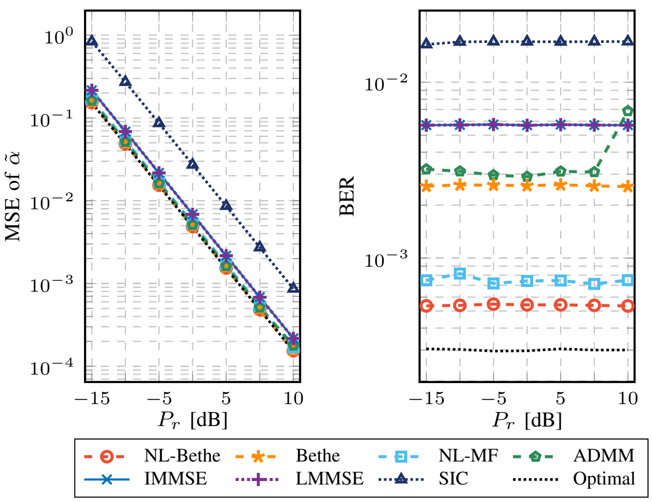



Fig. 7. MSE and BER performance of various methods with respect to the probing signal power $P_r$.

transceiver configuration still produce highly accurate results [38]. As evident from both figures, the convergence trends of $E_{\tilde{\alpha}}$ with and without noise are similar, with both cases converging around the 10-th iteration. Specifically, the MSE performance at convergence is comparable, as the estimation of $\tilde{\boldsymbol{\alpha}}$ in (38) exhibits robustness to noise variance. In contrast, for communication MSE $E_x$, noise learning enables convergence to an exceptionally low steady-state value, whereas its absence leads to slower convergence and higher residual errors. Overall, the results in both figures consistently demonstrate the significant advantages of noise learning in Bethe method.

**Remark 3.** In the simulations, we assume that $N_0$ is unknown in the case of Bethe method with noise learning, and $\delta$ is simply initialized to 1. Interestingly, the simulation results in Fig 2 and Fig. 3 indicate that updating the unknown noise variance in each iteration yields even better performance than using the exact noise variance. We attribute this phenomenon to the fact that the update formula as in (44) compensates for errors induced by the S&C interference and the system noise inherent to the approximate algorithm itself, rather than merely providing an estimate of the true noise variance.

## VI. Experimental Results

In this section, we evaluate the performance of our proposed methods, the mean-field method with noise learning (NL-MF) and Bethe method with or without noise learning (denoted as Bethe and NL-Bethe, respectively), for joint radar parameter estimation and communication signal detection in ISAC systems. The numerical results are compared with several baseline methods, including the heuristic LMMSE (the globally linear optimal scheme) as defined in (14), iterative MMSE (IMMSE, the linear optimal iterative alternating optimization scheme), and successive interference cancellation (SIC, the linear optimal IC scheme). Specifically, the IMMSE method employs an iterative processing architecture, utilizing LMMSE receivers to alternately estimate the radar parameters and the communication signals within each iteration. In addition, SIC initially treats the communication signals as Gaussian interference, performing LMMSE estimation of the radar parameters. Subsequently, it subtracts the estimated sensing quantity from the received signals to detect the communication signals, leveraging the statistical information of the sensing errors.

Our comparisons focus on two key performance metrics: sensing accuracy, evaluated through the MSE of the estimated reflection coefficients, and communication performance, characterized by the bit error rate (BER). In subsection VI-A, numerical simulations are conducted across various ISAC system configurations, including correlated MIMO channels with diverse correlation coefficients and varying modulation orders. Furthermore, the computational complexity of different receivers will be analyzed in VI-B to validate the efficiency of our proposed methods.

TABLE II
COMPLEXITY COMPARISON OF DIFFERENT METHODS

| Method | Sensing | Communication | Noise learning |
|---|---|---|---|
| LMMSE | $\mathcal{O}(N_rL^2S + N_rKLS + KL^2S + K^2LS^2 + K^3S^2 + N_rKS^2)$ | | – |
| ADMM | $\mathcal{O}(N_rK^2 + N_rKS + (K^3 + K^2S + KS^2)T_A)$ | | – |
| SIC | $\mathcal{O}(N_rL^2S)$ | $\mathcal{O}(N_rK^2 + K^3 + N_rKS)$ | – |
| IMMSE | $\mathcal{O}(N_rL^2S + N_rLKS + KLST + N_rL^2T)$ | $\mathcal{O}(N_rK^2 + K^3 + N_rKST + N_rLST)$ | – |
| NL-MF | $\mathcal{O}(N_rL^2S + N_rLKS + KLST + N_rL^2T)$ | $\mathcal{O}(N_rK^2 + K^2ST + KLST + MKST)$ | $\mathcal{O}(N_rKST + N_rLST)$ |
| Bethe | $\mathcal{O}(N_rL^2S + N_rLKS + KLST + N_rL^2T)$ | $\mathcal{O}(N_rKST + N_rLST + MKST)$ | – |
| NL-Bethe | $\mathcal{O}(N_rL^2S + N_rLKS + KLST + N_rL^2T)$ | $\mathcal{O}(N_rKST + MKST)$ | $\mathcal{O}(N_rKST + N_rLST)$ |

### A. Numerical Results

The performance comparisons are conducted from three key perspectives: 1) MSE and BER performance as functions of $E_b/N_0$, 2) the trade-off between MSE and BER performance with respect to various sensing and communication power, and 3) MSE and BER performance versus the number of snapshots $S$. In all numerical simulations, we set the maximum iteration number $T$ of all the iterative methods (NL-Bethe, Bethe, NL-MF, IMMSE) to 10. The transmit antenna at BS $N_t$ is configured to 12.

*1) MSE and BER performance versus $E_b/N_0$:* Fig. 4 demonstrates the sensing and communication performance under 16-QAM with $\zeta_R = 0.05$, $\zeta_T = 0.05$, $N_t = 12$, and $S = 10$. The sensing power $P_r$ and communication signal power are kept constant. The ISAC configurations differ across Fig. 4a ($K = 8, L = 4, N_r = 18$) and Fig. 4b ($K = 32, L = 4, N_r = 48$). As illustrated, the proposed methods consistently achieve superior sensing accuracy and communication reliability compared to other methods. For sensing performance, the proposed NL-Bethe, Bethe, and NL-MF share similar MSE performance, achieving improvements of approximately 2.2 dB and 3.5 dB at an MSE level of $-29$ dB in Fig. 4a and Fig. 4b, respectively, when compared to LMMSE and IMMSE. In contrast, the performance of SIC is far from satisfactory. Regarding communication performance, the proposed NL-Bethe outperforms LMMSE and IMMSE with improvements of 2 dB in Fig. 4a and 3.8 dB in Fig. 4b at the BER of $1 \times 10^{-3}$.

Fig. 5 presents the performance of the proposed methods under different modulation orders (16-QAM and 64-QAM). The simulations assume moderate correlation coefficients as $\zeta_R = 0.1$, $\zeta_T = 0.1$ with $K = 12, L = 8, N_r = 28$, $N_t = 12$, and $S = 10$. The performance of NL-Bethe, Bethe and NL-MF exhibit high consistency across different modulation orders, surpassing LMMSE and IMMSE by 1.5-2.2 dB for both sensing and communication tasks.

*2) The trade-off between MSE and BER performance:* To evaluate the trade-off between MSE and BER performance of our proposed methods, we vary either the probing signal power or the communication signal power while holding the other constant to conduct comparisons among different methods in Fig. 6 and Fig. 7. The simulations are performed in correlated channels under 16-QAM with $K = 16, L = 8, N_r = 32$, $S = 15$, $\zeta_R = 0.05$ and $\zeta_T = 0.05$.

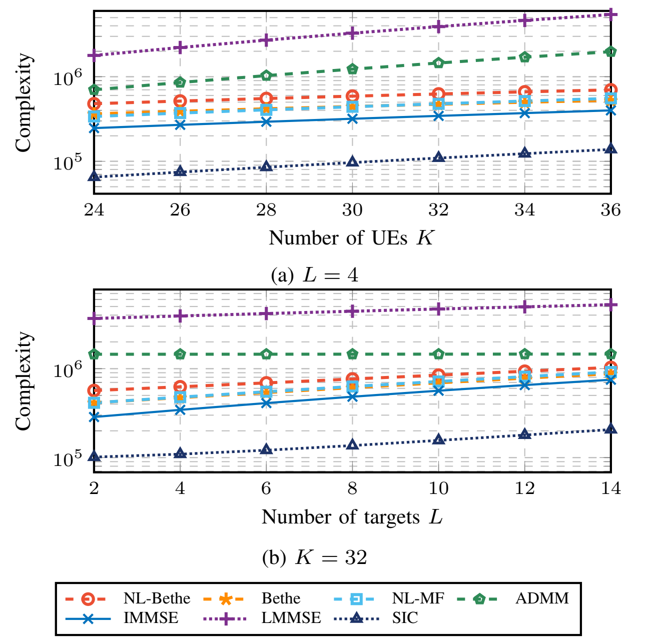



Fig. 9. Complexity comparisons of various methods with respect to (a) the number of UEs $K$ and (b) the number of targets $L$.

To further illustrate the effectiveness of our algorithms, we have included the alternating direction method of multipliers (ADMM)-based receiver proposed in [15], denoted as ADMM, along with theoretical performance bounds labeled as "Optimal" in the simulations. The parameters for ADMM have been optimized through Monte Carlo simulations, with the iteration number set to $T_A = 30$. In each MSE simulation, the theoretical bounds represent the MMSE estimation performance for the reflection coefficients without the influence of communication signals, while in the BER simulations, they represent the maximum likelihood (ML) detection performance without sensing interference.

The MSE and BER performance of various methods are compared with respect to the communication signal power $E_s$ with the probing signal power set to unity in Fig. 6. Over a relatively wide dynamic range of signal power variations, NL-Bethe, Bethe, and NL-MF can maintain consistent MSE performance. In contrast, the MSE of LMMSE and IMMSE exhibits an upward trend as the signal power $E_s$ increases, and the MSE performance of the SIC algorithm deteriorates significantly when the signal power is relatively high. Additionally, the MSE performance of ADMM is notably inferior at lower values of $E_s$. Regarding BER performance, the proposed NL-Bethe algorithm demonstrates an approximate 2 dB improvement over ADMM and a 2.5 dB enhancement compared to LMMSE and IMMSE when BER = $1 \times 10^{-4}$. In Fig. 7, the probing signal power $P_r$ is varied to examine the performance of various methods. It can be observed that NL-Bethe, Bethe, and NL-MF present better MSE performance with their BER performance remaining relatively invariant to changes in $P_r$. In conclusion, our proposed methods show robust performance across varying sensing and communication power resources, emphasizing their effectiveness in practical applications.

*3) MSE and BER performance versus the number of snapshots $S$:* Fig. 8 evaluates the impact of the number of snapshots $S$ on the performance of the proposed methods. In this simulation, we set $N_r = 32, K = 16$, and $L = 8$ with 16-QAM transmission. Increasing the number of snapshots improves both sensing and communication performance, as shown by the steady decline in MSE and BER for all methods. The NL-Bethe and NL-MF methods achieve lower MSE and BER than other baseline methods. The results demonstrate the consistent efficiency of the proposed methods in leveraging limited sensing data under varying snapshot numbers.

### B. Complexity Analysis

A comparative analysis of computational requirements of various methods is summarized in Table II. Specifically, we compare our proposed NL-MF, Bethe and NL-Bethe methods against LMMSE, ADMM, SIC, and IMMSE. The overall complexity assessment can be divided into three components: sensing, communication, and noise learning.

As listed in Table II, LMMSE takes $\mathcal{O}(K^2LS^2 + K^2LS + K^3S^2)$ operations for the matrix inversion in (14), and the complexity order of ADMM can be referenced in [15]. IMMSE, NL-MF, Bethe and NL-Bethe share the same sensing complexity. With respect to communication tasks, NL-MF necessitates $\mathcal{O}(N_rK^2+K^2ST+KLST+MKST)$ operations to update $\mathbf{x}$. In comparison, the computational complexity of NL-Bethe is at the order of $\mathcal{O}(N_rKST + N_rST + MKST)$. Furthermore, the computation operations required for the noise variance updates in both NL-MF and NL-Bethe are of the order $\mathcal{O}(N_rKST + N_rLST)$.

Fig. 9 further illustrates the complexity comparisons of various methods versus $K$ and $L$, where the complexity is quantified by the number of multiplication operations. For both cases, the parameters are fixed as $N_r = 48$, $N_t = 12$, $S = 10$, $T = 10$, and $M = 16$. As illustrated in Fig. 9a ($L = 4$), with the increase in the number of UEs $K$, LMMSE and ADMM exhibit a relatively rapid growth rate in computational complexity among all schemes, whereas the other methods demonstrate comparatively moderate growth rates. In Fig. 9b ($K = 32$), LMMSE maintains the highest complexity, and the computational complexity of ADMM lies between that of LMMSE and NL-Bethe. When the number of sensing targets $L$ increases, the computational complexity of NL-Bethe, Bethe, NL-MF, and IMMSE follows similar trends, while SIC consistently achieves the lowest complexity.

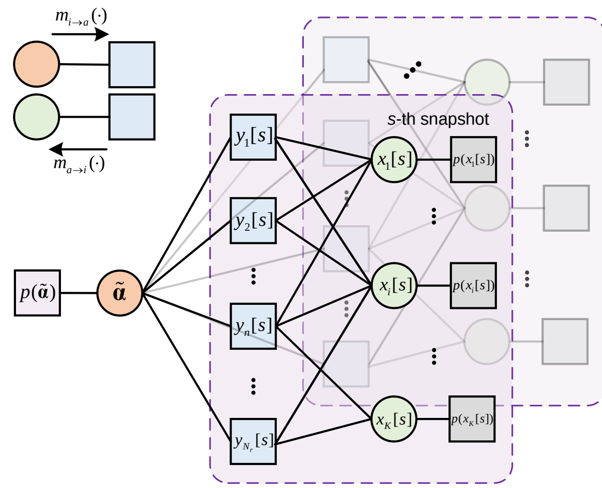



Fig. 10. The factor graph representation of the uplink ISAC receiver.

Additionally, the complexity of NL-Bethe is slightly higher than that of IMMSE, NL-MF, and Bethe in both simulation settings. In summary, LMMSE exhibits the highest computational complexity, while IMMSE, NL-MF and NL-Bethe possess comparable complexity levels.

## VII. Conclusion

In this work, we developed two variational Bayesian message passing methods for joint radar parameter estimation and communication signal detection in uplink ISAC systems. We derived closed-form expressions for the mean-field and Bethe variational free energies associated with the uplink ISAC system model, leading to the corresponding message passing equations. Additionally, noise variance learning methods were also introduced to further enhance performance. To validate the effectiveness of the proposed Bethe method, we provided the SE analysis to evaluate sensing and communication performance. Numerical simulations across various ISAC configurations show that both the mean-field and Bethe methods consistently outperform baseline methods in terms of MSE and BER. Furthermore, the computational complexity analysis confirms that the proposed methods achieve superior performance with comparable computational efficiency.

## Appendix A
## Mean-field Free Energy Derivation

Bringing the variational distribution (22) into (15), the mean-field free energy can be expressed as

$$\begin{aligned}\mathcal{F}_{\mathrm{MF}}(q) &= \mathbb{E}_{q(\mathbf{x},\tilde{\boldsymbol{\alpha}})}[\ln p(\mathbf{x},\tilde{\boldsymbol{\alpha}},\mathbf{y})] + \mathbb{E}_{q(\mathbf{x},\tilde{\boldsymbol{\alpha}})}[\ln q(\mathbf{x},\tilde{\boldsymbol{\alpha}})] \\ &= -\mathbb{E}_{q(\mathbf{x},\tilde{\boldsymbol{\alpha}})}[\ln p(\mathbf{y}|\mathbf{x},\tilde{\boldsymbol{\alpha}})] + \mathrm{KL}(q(\mathbf{x},\tilde{\boldsymbol{\alpha}})\|p(\mathbf{x})p(\tilde{\boldsymbol{\alpha}})).\end{aligned} \tag{55}$$

$$\begin{aligned}\sum_{ia} \ln Z^{i\to a} &= \sum_{ia} \ln \int p(x_i) \exp\{-|x_i|^2 \sum_b A_{b\to i} + 2\Re\{\sum_b B^*_{b\to i} x_i\}\} \exp\{|x_i|^2 A_{a\to i} - 2\Re\{B^*_{a\to i} x_i\}\} \\ &\overset{(a)}{\approx} \sum_{ia} \{\ln \tilde{Z}^i + \ln[1 + \underbrace{(|\hat{x}_i|^2 + \sigma^2_{\hat{x}_i}) A_{a\to i} - 2\Re\{\hat{x}_i B^*_{a\to i}\} + \sigma^2_{\hat{x}_i} |B_{a\to i}|^2}_{\Delta}]\} \\ &\overset{(b)}{\approx} \sum_{ia} \ln \tilde{Z}^i + \frac{|\hat{x}_i|^2 - 2\Re\{\hat{x}_i z_i^*\} + \sigma^2_{\hat{x}_i}}{\xi_i} + \sigma^2_{\hat{x}_i} \sum_a |B_{a\to i}|^2. \end{aligned} \tag{59}$$

The term $\mathbb{E}_{q(\mathbf{x},\tilde{\boldsymbol{\alpha}})}[p(\mathbf{y}|\mathbf{x},\tilde{\boldsymbol{\alpha}})]$ is given as

$$\begin{aligned}&\mathbb{E}_{q(\mathbf{x},\tilde{\boldsymbol{\alpha}})}[\ln p(\mathbf{y}|\mathbf{x},\tilde{\boldsymbol{\alpha}})] = \sum_s \mathbb{E}_{q(\mathbf{x}[s])q(\tilde{\boldsymbol{\alpha}})}[\ln p(\mathbf{y}[s]|\mathbf{x}[s],\tilde{\boldsymbol{\alpha}})] \\ &= \sum_s \mathbb{E}_{q(\mathbf{x}[s])q(\tilde{\boldsymbol{\alpha}})}[-\frac{\|\mathbf{y}[s] - \mathbf{H}_c\mathbf{x}[s] - \mathbf{G}_s\tilde{\boldsymbol{\alpha}}\|^2}{\delta[s]}] - \ln|\pi\delta[s]|^{N_r} \\ &= -\sum_s \frac{1}{\delta[s]}[\|\mathbf{y}[s] - \mathbf{H}_c\hat{\mathbf{x}}[s] - \mathbf{G}_s\hat{\tilde{\boldsymbol{\alpha}}}\|^2 + \mathrm{Tr}\{\mathbf{H}_c\boldsymbol{\Sigma}_{\hat{\mathbf{x}}[s]}\mathbf{H}_c^H\} \\ &+ \mathrm{Tr}\{\mathbf{G}_s\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}\mathbf{G}_s^H\} + N_r\ln(\pi\delta[s])],\end{aligned} \tag{56}$$

where $\boldsymbol{\Sigma}_{\hat{\mathbf{x}}[s]} = \mathrm{diag}([\sigma^2_{\hat{x}_1[s]},...,\sigma^2_{\hat{x}_K[s]}]^T)$, and $\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}$ is given in (25).

To derive the expression of KL divergence, we first reformulate (23) as

$$\begin{aligned}q(\tilde{\boldsymbol{\alpha}}) \propto & p(\tilde{\boldsymbol{\alpha}}) \exp\{-(\tilde{\boldsymbol{\alpha}}^H - \boldsymbol{\mu}^H_{\tilde{\boldsymbol{\alpha}}})\boldsymbol{\Sigma}^{-1}_{\tilde{\boldsymbol{\alpha}}}(\tilde{\boldsymbol{\alpha}} - \boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}})\} \\ \propto & p(\tilde{\boldsymbol{\alpha}}) \prod_s \exp\{-\mathbb{E}_{q\backslash\tilde{\alpha}}[\frac{1}{N_0}\|\mathbf{y}[s] - \mathbf{H}_c\mathbf{x}[s] - \mathbf{G}_s\tilde{\boldsymbol{\alpha}}\|^2]\} \\ \propto & p(\tilde{\boldsymbol{\alpha}}) \prod_s \exp\{-(\tilde{\boldsymbol{\alpha}}^H - \boldsymbol{\mu}^H_{\tilde{\boldsymbol{\alpha}}}[s])\boldsymbol{\Sigma}^{-1}_{\tilde{\boldsymbol{\alpha}}}[s](\tilde{\boldsymbol{\alpha}} - \boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}[s])\},\end{aligned} \tag{57}$$

where $\boldsymbol{\Sigma}^{-1}_{\tilde{\boldsymbol{\alpha}}}[s] = \frac{1}{N_0}\mathbf{G}_s^H\mathbf{G}_s$, $\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}[s] = \boldsymbol{\Sigma}_{\tilde{\boldsymbol{\alpha}}}[s]\mathbf{G}_s^H(\mathbf{y}[s] - \mathbf{H}_c\hat{\mathbf{x}}[s])$, $\boldsymbol{\Sigma}^{-1}_{\tilde{\boldsymbol{\alpha}}} = \frac{1}{N_0}\mathbf{G}^H\mathbf{G}$, and $\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}} = \boldsymbol{\Sigma}_{\tilde{\boldsymbol{\alpha}}}\mathbf{G}^H(\mathbf{y} - \mathbf{H}\hat{\mathbf{x}})$. And the term $\mathrm{KL}(q(\mathbf{x},\tilde{\boldsymbol{\alpha}})\|p(\mathbf{x})p(\tilde{\boldsymbol{\alpha}}))$ can be calculated as

$$\begin{aligned}&\mathrm{KL}(q(\mathbf{x},\tilde{\boldsymbol{\alpha}})\|p(\mathbf{x})p(\tilde{\boldsymbol{\alpha}})) = \mathbb{E}_{q(\mathbf{x})q(\tilde{\boldsymbol{\alpha}})}[\ln\frac{\prod_s q(\mathbf{x}[s])q(\tilde{\boldsymbol{\alpha}})}{\prod_s p(\mathbf{x}[s])p(\tilde{\boldsymbol{\alpha}})}] \\ &= \sum_s\Big\{\sum_i[\mathbb{E}_{q(\mathbf{x}[s])}[-\frac{|x_i[s] - z_i[s]|^2}{\xi_i[s]}] - \ln Z(z_i[s],\xi_i[s]) \\ &+ \mathbb{E}_{q(\tilde{\boldsymbol{\alpha}})}[-(\tilde{\boldsymbol{\alpha}}^H - \boldsymbol{\mu}^H_{\tilde{\boldsymbol{\alpha}}}[s])\boldsymbol{\Sigma}^{-1}_{\tilde{\boldsymbol{\alpha}}}[s](\tilde{\boldsymbol{\alpha}} - \boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}[s])]\Big\} \\ &- \ln Z(\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}},\boldsymbol{\Sigma}_{\tilde{\boldsymbol{\alpha}}}) \\ &= \sum_s\Big\{\sum_i[-\frac{|\hat{x}_i[s] - z_i[s]|^2 + \sigma^2_{\hat{x}_i[s]}}{\xi_i[s]} - \ln Z(z_i[s],\xi_i[s])] \\ &- \mathrm{Tr}\{\boldsymbol{\Sigma}^{-1}_{\tilde{\boldsymbol{\alpha}}}[s][\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}} + (\hat{\tilde{\boldsymbol{\alpha}}} - \boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}[s])(\hat{\tilde{\boldsymbol{\alpha}}}^H - \boldsymbol{\mu}^H_{\tilde{\boldsymbol{\alpha}}}[s])]\}\Big\} \\ &- \ln Z(\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}},\boldsymbol{\Sigma}_{\tilde{\boldsymbol{\alpha}}}).\end{aligned} \tag{58}$$

Combine (56) and (58), the mean-field free energy can then be obtained as in (30) and (31).

## Appendix B
## Bethe Free Energy Derivation

Fig. 10 illustrates the factor graph representation of the joint posterior distribution of the reflection coefficients and communication signals $p(\mathbf{x},\tilde{\boldsymbol{\alpha}}|\mathbf{y})$. The factor nodes are categorized into three types: observation nodes $y_n[s]$, the prior for reflection coefficients $p(\tilde{\boldsymbol{\alpha}})$, and the prior for communication signals $p(x_i[s])$. The variable nodes comprise the vector $\tilde{\boldsymbol{\alpha}}$ and scalars $x_i[s]$. Messages are transmitted bidirectionally between the observation nodes and all variable nodes, while the priors $p(\tilde{\boldsymbol{\alpha}})$ and $p(x_i[s])$ send unidirectional messages to their corresponding variables. Let $i$ index all variable nodes, and $a$ index observation factor nodes. We define $m_{a\to i}(\cdot)$ as the normalized message from observation node $a$ to variable node $i$, and $m_{i\to a}(\cdot)$ as the reverse message. Denote $f_a(\cdot)$ as the likelihood function of the $a$-th observation factor node, where the arguments represent all variables connected to it.

With a slight abuse of notation, we now use $x_i$ to represent either the communication signal $x_i[s]$ or the reflection coefficient vector $\tilde{\boldsymbol{\alpha}}$. Following [37], we define the intermediate parameters as

$$V_a = \sum_i |C_{ai}|^2\sigma^2_{i\to a},\ \omega_a = \sum_i C_{ai}\hat{x}_{i\to a}, \tag{60}$$

where $\sigma^2_{i\to a} = \int \mathrm{d}x_i\, |x_i|^2\, m_{i\to a}(x_i) - |\hat{x}_{i\to a}|^2, \hat{x}_{i\to a} = \int \mathrm{d}x_i\, x_i m_{i\to a}(x_i)$. Here, if $i$ indexes a communication signal, then $C_{ai} = h_{ni}$ with $a = n + (s-1)N_r$. When $i$ refers to the reflection coefficient vector, $C_{ai}$ represents the $a$-th row of $\mathbf{G}$, denoted by $\mathbf{g}_a$. We further define

$$A_{a\to i} = \frac{|C_{ai}|^2}{N_0 + \sum_{j\neq i}|C_{aj}|^2\sigma^2_{j\to a}} \overset{(a)}{\approx} \frac{|C_{ai}|^2}{N_0 + V_a}, \tag{61}$$

$$B_{a\to i} = \frac{C^*_{ai}(y_a - \sum_{j\neq i} C_{aj}\hat{x}_{j\to a})}{N_0 + \sum_{j\neq i}|C_{aj}|^2\sigma^2_{j\to a}} \overset{(b)}{\approx} \frac{C^*_{ai}(y_a - \omega_a)}{N_0 + V_a}, \tag{62}$$

where steps (a) and (b) involve omitting the lower-order terms $|C_{ai}|^2\sigma^2_{i\to a}$ and $C_{aj}\hat{x}_{j\to a}$, respectively. Given (60)-(62), the normalized messages $m_{a\to i}(x_i)$, $m_{i\to a}(x_i)$ and $q(x_i)$ can be formulated as [37]

$$m_{a\to i}(x_i) = \frac{\exp\{-|x_i|^2 A_{a\to i} + 2\Re\{B^*_{a\to i}x_i\}\}}{Z^{a\to i}}, \tag{63}$$

$$m_{i\to a}(x_i) = \frac{p(x_i)}{Z^{i\to a}}\exp\{-|x_i|^2\sum_{b\neq a} A_{b\to i} + 2\Re\{\sum_{b\neq a} B^*_{b\to i}x_i\}\}, \tag{64}$$

$$q(x_i) = \frac{p(x_i)}{\tilde{Z}^i}\exp\{-|x_i|^2\sum_a A_{a\to i} + 2\Re\{\sum_a B^*_{a\to i}x_i\}\}. \tag{65}$$

Following [39] and [28], the Bethe free energy (21) as a function of BP fixed-point messages can be reformulated as

$$\mathcal{F}_{\text{Bethe}}(q) = -\sum_a \ln Z^a - \sum_{ia}(\ln\tilde{Z}^i - \ln Z^{i\to a}) - \sum_i \ln\tilde{Z}^i, \tag{66}$$

where $Z^a = \int \mathrm{d}\mathbf{x}_a f_a(\mathbf{x}_a)\prod_i m_{i\to a}(x_i)$. The calculations of

$\sum_i \ln \tilde{Z}^i$ and $\sum_a \ln Z^a$ are straightforward, while the key approximation technique lies in computing $\sum_{ai} \ln Z^{a\to i}$.

First, we consider the case where $i$ indexes the communication signals. The term $\sum_{ia} \ln Z^{a\to i}$ can be calculated as in (59), where $\xi_i = \frac{1}{\sum_a A_{a\to i}}$, $z_i = \frac{\sum_a B_{a\to i}}{\sum_a A_{a\to i}}$. In step (a) of (59), we perform a second-order Taylor expansion of $\exp\{|x_i|^2 A_{a\to i} - 2\Re\{B^*_{a\to i} x_i\}\}$ at $x_i = \hat{x}_i = \mathbb{E}_{q(x_i)}[x_i]$. Step (b) in (59) applies a first-order Taylor expansion of $\ln(1+\Delta)$ with respect to the term $\Delta$. Similarly, if $i$ indexes the reflection coefficient vector, we perform a second-order Taylor expansion for the multivariate exponent function $\exp\{\frac{\tilde{\boldsymbol{\alpha}}^H(\mathbf{g}_a^H\mathbf{g}_a)\tilde{\boldsymbol{\alpha}}}{N_0+V_a} - 2\Re\{\frac{y_a-\omega_a}{N_0+V_a}\mathbf{g}_a^H\tilde{\boldsymbol{\alpha}}\}\}$ at $\tilde{\boldsymbol{\alpha}} = \hat{\tilde{\boldsymbol{\alpha}}} = \mathbb{E}_{q(\tilde{\boldsymbol{\alpha}})}[\tilde{\boldsymbol{\alpha}}]$. Subsequently, a corresponding first-order Taylor expansion of the logarithmic term is applied to obtain $\sum_{ia} \ln Z^{a\to i}$.

By defining $\sum_a \mathcal{F}_a = \sum_a \ln Z^a$, $\sum_{ia} \mathcal{F}_{ia} = \sum_{ia}\left(\ln \tilde{Z}^i - \ln Z^{i\to a}\right)$, and $\sum_i \mathcal{F}_i = \sum_i \ln \tilde{Z}^i$, we arrive at the free energy expression (66) in the context of our system model with each term expressed as

$$\sum_a \mathcal{F}_a = \sum_a \ln[\frac{1}{\pi(N_0+V_a)}\exp\{-\frac{|y_a-\omega_a|^2}{N_0+V_a}\}]. \quad (67)$$

$$\begin{aligned}\sum_i \mathcal{F}_i &= \sum_{i,s} \ln Z(z_i[s],\xi_i[s]) + \frac{|z_i[s]|^2}{\xi_i[s]}\\ &+ \ln Z(\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}},\boldsymbol{\Sigma}_{\boldsymbol{\mu}}) + \boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}^H\boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{-1}\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}, \quad (68)\\ \sum_{ia}\mathcal{F}_{ia} &= \mathrm{Tr}\{\boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{-1}(\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}} + \hat{\tilde{\boldsymbol{\alpha}}}\hat{\tilde{\boldsymbol{\alpha}}}^H)\} - 2\Re\{\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}^H\boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{-1}\hat{\tilde{\boldsymbol{\alpha}}}\}\\ &+ \sum_{i,s}\frac{|\hat{x}_i[s]|^2 - 2\Re\{\hat{x}_i[s]z_i^*[s]\} + \sigma^2_{\hat{x}_i[s]}}{\xi_i[s]}\\ &+ \sum_a \frac{|\omega_a - \mathbf{g}_a\hat{\tilde{\boldsymbol{\alpha}}}|^2}{V_a} + \sum_a \frac{|\omega_a - \sum_i h_{ni}\hat{x}_i[s]|^2}{V_a}, \quad (69)\end{aligned}$$

where the definitions of $\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}, \boldsymbol{\Sigma}_{\boldsymbol{\mu}}$ follow the same form as $z_i[s]$, $\xi_i[s]$. Specifically, $V_a$ can be simplified as

$$V_a = \sum_i |h_{ni}|^2\sigma^2_{\hat{x}_i[s]} + \mathbf{g}_a^H\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}}\mathbf{g}_a. \quad (70)$$

By calculating the stationary point of free energy (66), the fixed-point solution of $\omega_a$ is given by

$$\omega_a = (1+\frac{V_a}{N_0})(\sum_i h_{ni}\hat{x}_i[s] + \mathbf{g}_a\hat{\tilde{\boldsymbol{\alpha}}}) - \frac{V_a y_a}{N_0}. \quad (71)$$

In (67)-(71), the index $a$ equals to $n+(s-1)N_r$ with $s = 1,2,...,S$. Substituting the fixed-point values of $V_a$ and $\omega_a$ into (66) and replacing $N_0$ with $\delta[s]$, we derive the Bethe free energy expression as given in (32) with

$$\begin{aligned}\mathrm{KL}^*(q\|p_0) &= -\sum_i \mathcal{F}_i - \sum_{i,s}\frac{|\hat{x}_i[s]|^2 - 2\Re\{\hat{x}_i[s]z_i^*[s]\} + \sigma^2_{\hat{x}_i[s]}}{\xi_i[s]}\\ &- \mathrm{Tr}\{\boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{-1}(\boldsymbol{\Sigma}_{\hat{\tilde{\boldsymbol{\alpha}}}} + \hat{\tilde{\boldsymbol{\alpha}}}\hat{\tilde{\boldsymbol{\alpha}}}^H)\} + 2\Re\{\boldsymbol{\mu}_{\tilde{\boldsymbol{\alpha}}}^H\boldsymbol{\Sigma}_{\boldsymbol{\mu}}^{-1}\hat{\tilde{\boldsymbol{\alpha}}}\}. \quad (72)\end{aligned}$$

Notably, (30) and (32) can be interpreted as extensions of the mean-field free energy expression in [28, (10)] and the Bethe free energy expression in [28, (24)], respectively. While the formulations in [28] are based on fully factorized scalar variables, our approach provides a more general framework by accommodating variable nodes in both vector form (e.g., $\tilde{\boldsymbol{\alpha}}$) and scalar form (e.g., $x_i[s]$).

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